Deep holes of projective Reed–Solomon codes beyond redundancy four: exact classifications and stable polar components

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Abstract

We classify, up to projective-semilinear equivalence, the deep-hole syndrome directions of projective Reed–Solomon codes at redundancies five and six over every prime power $q \geq 7$, with exact projective counts. At redundancy seven we give the complete split-free classification over the same range; it is a deep-hole classification whenever the covering radius is six, in particular for $q \geq 11$. The redundancy-five and six results extend the complete classifications previously known through redundancy four.

At redundancy five the list contains, besides the tangent and conjugate-secant families, an osculating-pair family selected by $q \bmod 3$, a characteristic-three nucleus point and wild Artin–Schreier family, and sporadic orbits exactly for $q \in \{7, 8, 9, 11, 13, 17, 19\}$. A syndrome determines a Hankel linear system of binary forms and is split-free precisely when the system contains no squarefree form that splits completely over $\mathbb{F}_q$. Here the system is a pencil of cubics: cyclic descent produces the arithmetic families, an integral genus-one fiber square controls the S_3 case, and exact certificate computations settle the remaining fields. The balanced $q = 8$ row gives 1116 relative $[10, 5, 6]_8$ MDS extensions and hence minimum-support AME$(10, 8)$ states and $[[9, 1, 5]]_8$ quantum MDS codes.

For higher redundancies we introduce coherent polar flags. A contraction removes a prospective root; retaining it as a forbidden marker is exactly what prevents a lower squarefree witness from lifting with a double root. The reduced contained branch of this induction admits an all-level irreducible-component exclusion theorem: a catalecticant-rowspace reduction and an integral Grassmannian calculation show that every recursively pointed contained component lies in the persistent or modular locus. Together with the transverse point bound, this implies that every split-free syndrome at every redundancy $r \geq 6$ lies in these loci once

$$q \geq 6r - 15 + \lfloor 2\sqrt{6r - 17} \rfloor.$$

Seroussi–Roth nonextendability together with Dür's completeness–covering-radius equivalence supplies radius $r - 1$ throughout this range. Thus these split-free directions are precisely the projective deep-hole directions. If $\operatorname{char} \mathbb{F}_q > r - 1$, the modular locus is empty, and the deep-hole directions are exactly the persistent tangent and conjugate-secant families.

Index Terms—Absolutely maximally entangled states, algebraic coding theory, deep holes, finite fields, maximum-distance-separable extensions, normal rational curves, projective Reed–Solomon codes, quantum MDS codes.

1 Introduction

This paper gives two complete all-field deep-hole classifications and one all-level component theorem beyond the known cases through redundancy four. At redundancies five and six we classify the projective deep-hole directions for every prime power $q \geq 7$, with exact projective counts and semilinear orbit laws. At redundancy seven we classify every split-free syndrome over the same range; this is a deep-hole classification whenever the covering radius is six, in particular for $q \geq 11$. The common mechanism is a coherent polar flag, a marked contraction that propagates splitting information without forgetting the factor needed for a squarefree lift. At arbitrary redundancy, a catalecticant-rowspace reduction and an integral bottom-component calculation then prove that every recursively pointed contained component is persistent or modular.

The mechanism is the elementary identity

$$g \in W_{\iota_\lambda f} \iff \lambda g \in W_f, \qquad \lambda g \text{ squarefree} \iff \lambda \nmid g$$

when g is squarefree. Retaining λ is therefore the essential coherence condition: without the marker, a lower divisor through λ lifts with a double root, so the naïve induction is false. The $q = 19$ example in Section 6 makes the failure concrete: an individual lower fiber appears exceptional, but its orbit contains no coherent consecutive-row polar line. Theorem 1.3 classifies the contained components, and Theorem 5.20 closes the transverse branch for

$$q \geq 6r - 15 + \left\lfloor 2\sqrt{6r - 17} \right\rfloor.$$

In this range Seroussi–Roth nonextendability together with Dür's completeness–covering-radius equivalence automatically gives radius $r - 1$. Thus the all-level conclusion is an unconditional containment theorem for projective deep-hole directions, not only for split-free syndrome directions.

Let $\mathrm{PRS}(k)$ denote the projective Reed–Solomon code of length $q+1$ obtained by evaluating homogeneous binary forms of degree $k - 1$ on $\mathbb{P}^1(\mathbb{F}_q)$. If the redundancy is

$$r = q + 1 - k,$$

a parity-check matrix has as columns the $\mathbb{F}_q$-points of the degree-$(r-1)$ normal rational curve

$$\mathcal{R}_{r-1} = \{(1, t, \ldots, t^{r-1}) : t \in \mathbb{F}_q\} \cup \{(0, \ldots, 0, 1)\} \subset \mathbb{P}^{r-1}.$$

Call a projective syndrome direction *split-free* if it is not contained in the span of $r-2$ distinct columns. When $\rho(\mathrm{PRS}(q+1-r)) = r - 1$, the split-free directions are precisely the projective syndrome directions of deep-hole cosets. The logical separation used throughout the paper is

$$\boxed{\text{split-free classification} \ + \ \rho(\mathrm{PRS}(q+1-r)) = r-1 \quad \Longrightarrow \quad \text{deep-hole classification.}}$$

The first term is proved geometrically; the covering-radius equality is a separate coding-theoretic input and is never inferred from a finite split-free census.

The correspondence between deep holes, one-symbol MDS extensions, and uncovered points of a normal rational curve was developed in [Kai17, Prop. 1] and [ZWK20, Lem. II.4 and Thms. I.4–I.6]. Zhang, Wan, and Kaipa classified projective Reed–Solomon deep-hole directions through redundancy four and identified redundancy five as the next case [ZWK20]. Theorem 1.1 closes that case; Theorem 1.2 gives the complete redundancy-six classification and the radius-delimited redundancy-seven result.

Theorem 1.1 (Complete redundancy-five classification)**.** Let $q \geq 7$ be a prime power. The covering radius of $\mathrm{PRS}(q-4)$ is 4. Its projective deep-hole syndromes are exactly the following disjoint families:

(i) the tangent family T, of size $q(q + 1)$;

(ii) the conjugate-secant family S, of size $q(q^2 - 1)/2$;

(iii) if $\operatorname{char}\mathbb{F}_q \neq 3$, the rational osculating-pair family O^+ when $q \equiv 2 \pmod 3$, of size $q(q+1)/2$, or the conjugate osculating-pair family O^- when $q \equiv 1 \pmod 3$, of size $q(q-1)/2$;

(iv) in characteristic three, one $\mathrm{PGL}_2(q)$-fixed nucleus point and one wild Artin–Schreier orbit W of size $(q^2-1)/2$;

(v) the sporadic $\mathrm{PGL}_2(q)$-orbits in Table 4, occurring exactly for $q \in \{7, 8, 9, 11, 13, 17, 19\}$.

There are no other deep-hole syndromes. The three corresponding nonsporadic totals are

$$\frac{q^2(q+3)}{2}, \qquad \frac{q(q+1)(q+2)}{2}, \qquad \frac{q^3+3q^2+q+1}{2},$$

according as $q \equiv 1 \pmod 3$ with characteristic not three, $q \equiv 2 \pmod 3$, or $\operatorname{char}\mathbb{F}_q = 3$.

The main mechanism is a characteristic-free Hankel dictionary. A syndrome at redundancy r annihilates a linear system of binary forms of degree $r-2$. It is split-free if and only if this system contains no completely split squarefree form; it is a deep-hole direction when, in addition, $\rho(\mathrm{PRS}(q+1-r)) = r-1$. At redundancy five the system is a pencil of cubics. Its induced degree-three map of projective lines is either cyclic or has geometric monodromy S_3. Cyclic descent gives the arithmetic families in Theorem 1.1; in the S_3 case the off-diagonal fiber square is an integral curve of arithmetic genus one, and the singular-curve Weil bound [AP95, p. 468] forces a split fiber for $q \geq 23$. The certified finite computations in Section 7 handle the remaining fields.

Starting at redundancy six, the Hankel system is no longer a pencil. The useful replacement is the coherently parameterized line of contractions obtained by choosing one prospective root. Iteration produces a *polar flag*. Geometrically, naïve contraction keeps the lower divisor but forgets which point of $\mathbb{P}^1$ was removed. If the lower divisor passes through that point, putting the point back creates a double point rather than a squarefree divisor. The polar flag retains the removed point as a forbidden marker, exactly the datum needed for lifting. Thus the marked flag, rather than an independent collection of lower fibers, is the object on which induction is valid.

The resulting structural theorem has two outputs.

Theorem 1.2 (Redundancy-six deep holes and redundancy-seven split-free syndromes)**.** The following statements hold.

(a) For every prime power $q \geq 7$, the projective deep-hole directions of $\mathrm{PRS}(q-5)$ are exactly the persistent tangent and conjugate-secant families, the exceptional orbits listed for $q = 7, 8, 9, 11, 13$, and, when $q = 2^m$ with odd $m \geq 5$, one characteristic-two nucleus orbit. Here $m \geq 5$ avoids double counting: $m = 3$, or $q = 8$, is already the $|\mathcal{O}| = 9$ row of the exceptional table.

(b) The displayed finite domains through $q = 32$ have the certified split-free classification stated in Section 6. Together with the structural theorem, these finite classifications give the all-field result: for every $q \geq 13$ the split-free locus consists of the persistent families and, when $q = 2^m$ with m odd, one central nucleus point. This is a deep-hole classification only when $\rho(\mathrm{PRS}(q-6)) = 6$, in particular for $q \geq 11$.

At redundancy r, the semilinear orbits in the persistent conjugate-secant family are parameterized by

$$T/T^{r-1} \quad \text{modulo inversion and Frobenius,}$$

where $T = \{z \in \mathbb{F}_{q^2}^\times : z^{q+1} = 1\}$ is the norm-one torus and Frobenius is coefficient Frobenius. On the tangent family the unipotent coordinate transforms as $z \mapsto z + (r-1)u$; consequently its orbit decomposition changes precisely when $\operatorname{char}\mathbb{F}_q$ divides $r-1$.

The two parts have different scopes. Both are unconditional all-field syndrome classifications, but part (b) becomes a deep-hole classification only when $\rho(\mathrm{PRS}(q-6)) = 6$. Table 2 places the split-free result, radius gate, deep-hole conclusion, and remaining field gap in separate columns for each redundancy.

Theorem 1.3 (Stable components and the uniform transverse consequence)**.** The stable-component assertion $\mathrm{SC}(j)$ holds for every $j \geq 6$. In characteristic two, the modular component descended from the R5 cyclic plane is empty for every $r \geq 8$. Consequently, for every $r \geq 6$, if

$$q \geq 6r - 15 + \left\lfloor 2\sqrt{6r-17} \right\rfloor,$$

every split-free redundancy-r syndrome belongs to $\mathcal{P}_r \cup \mathcal{M}_r$. Calling these directions code deep holes requires the covering-radius equality $\rho(\mathrm{PRS}(q+1-r)) = r-1$; Seroussi–Roth nonextendability and Dür's equivalence supply it automatically under the displayed bound. Hence, in this range, the split-free directions are precisely the projective deep-hole directions and all lie in $\mathcal{P}_r \cup \mathcal{M}_r$.

Theorem 1.3 is the third principal contribution. Its proof is given in Section 5: products of the retained markers are dense in the full binary-form parameter space, so the bottom contractions close to the projectivized row space of a consecutive catalecticant. Irreducibility reduces containment to one bottom component, and each component is then treated explicitly. The same section gives an exact arithmetic test on the canonical three-dimensional section of every binary higher Lucas endpoint: for $d = 2^s$ over $\mathbb{F}_{2^m}$, that section contains a split squarefree degree-d form exactly when $s \mid m$. The equivalent projective-subline criterion was proved in the Lucas-subspace setting by [WWH26, Prop. 11]; our use identifies it inside the coherent Hankel endpoint. At the first fresh endpoint e_7, subspace polynomials show that the full PGL_2-orbit is shallow for every $m \geq 3$; this does not classify the remaining strata of the ambient Lucas carrier.

Relation to previous work. The tangent and conjugate-secant families and the coding/normal-rational curve dictionary come from [Kai17, Prop. 1] and [ZWK20, Lem. II.4 and Thms. I.4–I.6]; the covering-radius input in the high-rate range combines the Seroussi–Roth extension theorem [SR86, Thm. 1] with Dür's completeness–covering-radius equivalence [Dür94], as stated by Kaipa [Kai17, Sec. IV and Prop. 4(1)]. The broader arc/MDS and code-automorphism background is surveyed or established in [BL19, Dür87]. Earlier explicit-family results for projective and generalized projective Reed–Solomon codes include [ZW17, XHX17, Xu23]; the recent extension framework of [WDC23] further clarifies the deep-hole/MDS-extension interface. These works construct families, treat punctured or lower-redundancy regimes, or analyze the extension criterion; they do not supply the complete redundancy-five cubic-pencil exhaustion proved here. Classical apolarity, catalecticants, and binary-form rank strata are treated in [IK99, CS11]; our use of divided powers keeps the small-characteristic contraction formulas explicit. Kaipa, Patanker, and Pradhan classify the $\mathrm{PGL}_2(q)$-orbits of binary quartic forms and of lines relative to the twisted cubic [KPP25]. Their orbit classification is adjacent normal-form geometry for the redundancy-six problem; it does not impose the Hankel split-free condition or determine the corresponding deep-hole locus. The companion papers in this bundle treat three adjacent layers. Relative completion outside a prescribed conic gives the redundancy-three uncovered-syndrome problem [Rud26a]; the Clebsch rigidity paper asks the inverse question of recovering a non-GRS code from that locus [Rud26b]; and the conic-ideal quotient paper proves that secant matching products with the same endpoint divisor agree on the conic while their quotients retain 3-, 6-, and 10-dimensional A_3, B_3, H_3 factorization data, with quadratic sheet recovery and cubic orientation in the balanced cases [Rud26c]. That retained factorization memory is parallel to the forbidden marker here: both remember factor data erased by a natural reduction, but neither theorem is an input to the other. The MDS–CSS companion turns the balanced $q = 8$ extension row into the quantum consequences stated in Section 4 [Rud26d]. Apart from that explicit corollary, these companion results are comparisons rather than inputs to the classifications proved here. From the viewpoint of binary forms, the question is which linear systems contain a totally split squarefree member. Factorization statistics in linear families [CMP17] and normal-rational-curve nuclei [GH00, Lem. 1 and Thm. 1] are inputs. For splitting families, we use Wang's Theorem 3.6 [Wan26]: on the étale locus, the factorization type of a specialized polynomial is the cycle type of its Frobenius conjugacy class. The coding-theory references above identify deep syndromes with uncovered points of a normal rational curve and settle the lower-redundancy families; they do not classify the redundancy-five cubic pencils. The orbit-theoretic and factorization references supply normal forms and distribution tools, but not the coherent marked contraction used here. The nucleus results locate characteristic-dependent kernels, while the present work computes which of their coherent lifts meet the PRS splitting incidence. Accordingly, this paper's contributions are separated into the redundancy-five through seven classifications, the stable-component theorem, and the uniform transverse induction. Claim-by-claim novelty and dependency boundaries are recorded in the electronic ledger.

Organization. Section 2 first gives a proof-free guide to the argument, and Section 3 gives the Hankel dictionary. Section 4 proves Theorem 1.1. Sections 5 and 6 develop coherent polar induction and prove Theorem 1.2. Section 7 states the computational and formal trust boundary, and Section 8 records the exact open boundary.

2 Guide to the proof

This section is a proof-free guided tour. Its purpose is to install the information flow used throughout the paper before coordinates, finite exceptional tables, or formal trust boundaries intervene.

Syndrome to splitting system. A syndrome direction $[f] \in \mathbb{P}(\Gamma^{r-1}E)$ determines its Hankel annihilator $W_f \subset \mathrm{Sym}^{r-2} E^\vee$. A product of $r-2$ distinct rational linear factors lies in W_f exactly when f lies in the span of the corresponding normal-rational-curve columns. Thus the geometric classification begins with split-free systems. The separate covering-radius theorem is what promotes split-free directions to code deep holes; the dictionary alone does not.

The redundancy-five base cover. At redundancy five, W_f is a pencil of binary cubics. Its root incidence is a degree-three map of projective lines. The off-diagonal fiber square separates the cyclic cases, controlled by Frobenius on a deck transformation, from the S_3 case, controlled by rational points on a geometrically integral $(2, 2)$ curve. This is the bottom cover used at every later level.

Why independent lower fibers fail. At higher redundancy, choosing one lower witness forgets which factor was removed. If that lower witness contains the chosen factor, then multiplication restores it with multiplicity two rather than producing a squarefree form. The lost datum is therefore not auxiliary: it is the exact obstruction to lifting.

Coherent replacement and the fork. Choose a prospective root λ, contract f to $\iota_\lambda f$, and retain λ as a marked forbidden root. Iteration gives an ordered polar flag. Its first-polar line has two qualitatively different behaviors. If it is transverse to the lower bad locus and collision schemes, only finitely many parameters must be removed and a point count produces an escape. If it is scheme-theoretically contained, the point-count argument says nothing: ordinary catalecticant components and components created in small characteristic require a separate level-specific classification. Section 5 introduces the persistent and modular loci only when that distinction enters the proof.

Integral component closure. The ordered-root incidence of the bottom cubic pencil is a symmetric $(2, 2)$ form on a Grassmannian. Unique factorization leaves collision, one cyclic branch, and in characteristic three an anti-invariant diagonal collision. An integral Plücker-to-syndrome bridge identifies the cyclic branch with the coherent-Fano carrier, meaning the coefficient scheme on which the entire consecutive polar line lies in that carrier. After persistent saturation that carrier is empty, with characteristics two and three treated before residualization.

The all-level step is not formal marker substitution. Products of the retained markers are dense in the full binary-form parameter space, so their bottom contractions close to

$$\mathbb{P}(\operatorname{rowspace} \operatorname{Cat}_{m,4}(f)) .$$

This irreducible projective space lies in one component of the finite bottom bad union. The rank-two component is handled by the integral consecutive-catalecticant theorem, the tame Veronese contains no line, the characteristic-three cone is handled by its explicit ruling calculation, and the characteristic-two plane gives two overlapping zero blocks. Those blocks cover every coefficient from redundancy eight onward.

Lifting. On the transverse branch, a rational point of the lower ordered splitting cover outside the branch, diagonal, marker, and collision divisors produces a split squarefree lower witness avoiding all markers. Multiplying by the marked factors then lifts it squarefreely to W_f. Figure 1 records this sequence at the point where the construction is formalized.

Table 1: Mathematical dependency map.

Result	Printed geometric argument	Imported input
R5	Hankel dictionary; gcd strata; cyclic descent; integral $(2, 2)$ fiber square and point bound	lower tangent/secant families; high-rate radius theorem; singular-curve bound
R6	one-marker polar line; ordinary and binary contained components; transverse escape	R5 lower package and radius gate
R7	two-marker contraction; complete second-marker bad scheme, including cyclic and fixed-gcd collisions; contained rank-two theorem; transverse escape	R5/R6 lower packages; separate covering-radius result
All $r \geq 6$	catalecticant-rowspace transport; integral ordered-root incidence and factorization; Plücker bridge; coherent-Fano elimination; separate vertical fibres	R5 bottom cover and persistent/Lucas pullback identities

Table 2: Classification status by redundancy. "Unconditional" refers to the displayed field range. A split-free classification becomes a deep-hole classification only after the separate radius gate.

Red.	Field range	Split-free geometry	Deep holes/MDS extensions	Remaining gap
5	$q \geq 7$	Complete, unconditional	Complete, unconditional; $\rho = 4$	None
6	$q \geq 7$	Complete, unconditional	Complete, unconditional; $\rho = 5$	None
7	$q \geq 7$	Complete, unconditional	Complete for $q \geq 11$; conditional on $\rho = 6$ at $q = 7, 8, 9$	Radius at $q = 7, 8, 9$
$r \geq 6$	$q \geq Q_r$	Contained in $\mathcal{P}_r \cup \mathcal{M}_r$, unconditional	All split-free directions are deep holes; contained in $\mathcal{P}_r \cup \mathcal{M}_r$	Fields below Q_r; exact modular membership

3 The syndrome–Hankel dictionary

Let E be a two-dimensional vector space. We use divided powers for syndromes and ordinary symmetric powers for root forms. This avoids invalid binomial rescalings in small characteristic. For $f = (a_0 : \cdots : a_{r-1}) \in \mathbb{P}(\Gamma^{r-1}E)$, write $e_0, \ldots, e_{r-1}$ for the divided-power coordinate basis and $\nu(t) = (1, t, \ldots, t^{r-1})$, with $\nu(\infty) = e_{r-1}$. If x, y is a basis of E with dual basis X, Y, our perfect divided-power/symmetric-power pairing is

$$\left\langle x^{[n-i]}y^{[i]}, X^{n-j}Y^j \right\rangle = \delta_{ij}, \qquad \left\langle \sum_{i=0}^{n} a_i x^{[n-i]}y^{[i]}, \sum_{i=0}^{n} b_i X^{n-i}Y^i \right\rangle = \sum_{i=0}^{n} a_i b_i.$$

Thus the pairing is coefficient extraction, with no binomial factors in any characteristic. Define

$$H_f = \begin{pmatrix} a_0 & a_1 & \cdots & a_{r-2} \\ a_1 & a_2 & \cdots & a_{r-1} \end{pmatrix}, \qquad W_f = \ker H_f \subset \operatorname{Sym}^{r-2} E^\vee.$$

Table 3: Notation used from the Hankel dictionary onward.

Symbol	Meaning
q	field size; all numerical thresholds are thresholds on prime powers
k	code dimension in $\mathrm{PRS}(k)$
$r = q + 1 - k$	redundancy
$n = r - 1$	divided-power degree of a syndrome representative
E	a two-dimensional vector space over the current base field
$f \in \Gamma^n E$	divided-power representative of a projective syndrome
$W_f \subset \operatorname{Sym}^{n-1} E^\vee$	Hankel/apolar witness system annihilated by f
g, δ, κ	genus, deletion degree, and point-bound correction
d	degree of the unavailable marker scheme

Read $f = (a_0, \ldots, a_{r-1})$ as a finite sequence. A form $g \in W_f$ is the characteristic polynomial of a recurrence satisfied by that sequence, while a squarefree split g has a basis of geometric-sequence solutions. Thus split-free means that the syndrome sequence satisfies no order-$(r-2)$ recurrence with squarefree completely split characteristic polynomial.

Lemma 3.1 (Hankel criterion)**.** Let $\lambda_1, \ldots, \lambda_{r-2}$ be distinct points of $\mathbb{P}(E^\vee)(\mathbb{F}_q)$ and put $g = \prod_i \lambda_i$. Then

$$f \in \operatorname{span}\big(\nu(\lambda_1), \ldots, \nu(\lambda_{r-2})\big) \iff g \in W_f.$$

The same statement with repeated factors and Hasse derivatives gives the corresponding osculating spans. Both statements commute with $\mathrm{P\Gamma L}_2(q)$.

Proof. On an affine chart, the geometric sequences $\nu(\lambda_i)$ solve the recurrence with characteristic polynomial g; their independence is the Vandermonde determinant. A root at infinity gives the reversed-chart recurrence. For repeated roots, the confluent Vandermonde basis is formed by Hasse derivatives, so the same argument works in every characteristic. Equivariance follows because substitution transports both the normal rational curve and the recurrence. □

Corollary 3.2 (Split-free criterion)**.** Assume $\rho(\mathrm{PRS}(q+1-r)) = r-1$. A projective syndrome f is deep if and only if W_f contains no completely split squarefree form of degree $r-2$ over $\mathbb{F}_q$.

We use *split-free* without a covering-radius hypothesis and call a direction *shallow* when W_f contains a completely split squarefree form.

The first locus that persists under contraction is already visible here. If every member of W_f has a common quadratic factor, then that factor is either a rational square, an irreducible quadratic, or a product of two distinct rational factors. These are respectively the tangent deep family, the conjugate-secant deep family, and a shallow rational secant. The common factor cannot grow with redundancy. If H_f has rank one, then $\dim W_f = r-2$, which is incompatible with a common quadratic factor because the space of degree-$(r-2)$ forms divisible by that factor has dimension $r-3$. Otherwise $\dim W_f = r-3$, and

$$r-3 \le r-1-\deg\gcd(W_f),$$

so its degree is at most two.

3.1 Coding consequences and recognition

The dictionary has two immediate coding consequences. First, a split-free direction is an uncovered point of the normal rational curve and hence gives a one-column MDS extension whenever the stated covering-radius gate applies. The classification theorems therefore give exact projective deep-hole counts, not only normal forms. Second, they give recognition procedures. From a syndrome one forms H_f, computes W_f, and tests the listed geometric families: a common quadratic factor detects the persistent tangent and conjugate-secant families; the relevant contraction kernel detects the modular locus; and the finite exceptional fields are decided by the canonical orbit representatives printed in Table 4 and recorded in the supplement.

Recognition at the fixed redundancies proved here begins with row-reduction of H_f and the listed family-membership tests. Coherent polar induction replaces a direct search through the projective members of W_f.

4 Redundancy five: exceptional cubic covers

Here $f = (a_0 : \cdots : a_4)$ and W_f is the pencil annihilated by

$$\begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \end{pmatrix}.$$

These are divided-power coordinates; the coefficients of a cubic in W_f are ordinary symmetric-power coordinates. Factoring the formal quartic $\sum a_i T^i U^{4-i}$ would therefore be noninvariant in characteristics two and three.

Proposition 4.1 (Radius gate)**.** For every prime power $q \geq 7$,

$$\rho(\mathrm{PRS}(q-4)) = 4.$$

Proof. The coding/arc criterion identifies radius four with completeness of the quartic normal rational curve. The Seroussi–Roth extension theorem [SR86, Thm. 1], applied to the dual five-dimensional GRS code, gives nonextendability when

$$5 \leq \left\lfloor \frac{q}{2} \right\rfloor + 2,$$

apart from its even-field dimension-three exception. The inequality holds for $q \geq 7$, and the exception does not occur. Dür's completeness–covering-radius equivalence [Dür94], as stated by Kaipa [Kai17, Sec. IV and Prop. 4(1)], now gives radius four. Thus Corollary 3.2 applies. The finite certificate independently checks the same conclusion only in its declared finite domain; it is not the all-field radius proof. □

4.1 Gcd strata

Proposition 4.2 (Quadratic gcd)**.** If $\deg \gcd(W_f) = 2$, then

$$W_f = Q\langle T, U\rangle.$$

If Q is split squarefree, f is shallow. If Q is irreducible, f belongs to the conjugate-secant family; if $Q = \lambda^2$, it belongs to the tangent family. Conversely every point in those families has the displayed gcd. Their deep-family sizes are

$$|\mathrm{T}| = q(q+1), \qquad |\mathrm{S}| = \frac{q(q^2-1)}{2}.$$

Proof. The equality follows by dimension. A split Q and a linear cofactor avoiding its two roots give a split squarefree cubic. An irreducible Q survives in every member, while a square Q forces a repeated root in every member. Lemma 3.1, including its confluent and conjugate-root forms, identifies the corresponding syndromes with the secant and tangent spans. The sizes are the established tangent/conjugate-secant counts from the lower-redundancy classification [Kai17, Prop. 1] and [ZWK20, Thms. I.4–I.6]. □

Proposition 4.3 (Linear gcd)**.** Suppose $\deg \gcd(W_f) = 1$. Then f is shallow for $q \geq 8$. At $q = 7$ the same conclusion is part of Certificate R5.

Proof. Write $W_f = \lambda_0 V$ with V a basepoint-free quadratic pencil. If its degree-two map $\psi : \mathbb{P}^1 \to \mathbb{P}^1$ is separable, its rational deck involution ι gives the graph

$$\Gamma_\iota(\mathbb{F}_q) = \{(t, \iota(t)) : t \in \mathbb{P}^1(\mathbb{F}_q)\},$$

which has exactly $q+1$ rational points. A safe union bound deletes at most two fixed graph points, at most four graph points over the branch fibers, and at most the two graph points $(\lambda_0, \iota(\lambda_0))$ and $(\iota(\lambda_0), \lambda_0)$. Thus fewer than $q+1$ graph points are deleted when $q \geq 8$. A surviving point gives a split quadratic with two distinct roots avoiding λ_0, and hence a split squarefree cubic in W_f. In characteristic two an inseparable quadratic pencil is equivalent to $\langle T^2, U^2\rangle$; the two overlapping Hankel equations then force f onto the normal rational curve, contrary to the rank-two pencil hypothesis. □

4.2 The trivial-gcd cubic cover

Choose a basis c_1, c_2 of a trivial-gcd pencil and put

$$X_f = \{([u:v], t) : uc_1(t) + vc_2(t) = 0\} \subset \mathbb{P}^1 \times \mathbb{P}^1.$$

Proposition 4.4 (Cubic incidence and off-diagonal fiber square)**.** The curve X_f is geometrically integral of bidegree $(1, 3)$ and arithmetic genus zero. Projection to the root coordinate is birational, while projection to the pencil coordinate gives a degree-three morphism

$$\phi_f : \mathbb{P}^1 \longrightarrow \mathbb{P}^1.$$

If ϕ_f is separable, division of

$$c_1(t_1)c_2(t_2) - c_2(t_1)c_1(t_2)$$

by the diagonal bracket defines a residual curve $Y_f \subset \mathbb{P}^1 \times \mathbb{P}^1$ of bidegree $(2, 2)$ and arithmetic genus one. Its geometric components are the geometric-monodromy orbits on ordered pairs of distinct roots.

Proof. A component of X_f of bidegree $(0, e)$ would be a common factor of c_1, c_2; hence none exists. The bidegree genus formula gives genus zero. A root determines the unique pencil member containing it, so the root projection is birational. For a separable map the diagonal factor in the fiber-square determinant has multiplicity one. Removing its bidegree $(1, 1)$ from the fiber square leaves bidegree $(2, 2)$, whose arithmetic genus is one. The standard fiber-square/monodromy correspondence identifies its components with orbits on distinct ordered sheets. $\square$

Lemma 4.5 (Cyclic stratum)**.** Suppose ϕ_f is geometrically cyclic.

(i) In characteristic different from three there are two ramification points. If they are rational, the corresponding osculating-plane intersection is deep exactly when $q \equiv 2 \pmod 3$; if they are conjugate, it is deep exactly when $q \equiv 1 \pmod 3$.

(ii) In characteristic three all osculating planes meet in $n = (0 : 0 : 1 : 0 : 0)$, whose pencil is $\langle T^3, U^3 \rangle$; this fixed point is deep.

(iii) The remaining wild characteristic-three pencils have the form $\langle U^3, T^3 + aTU^2 \rangle$. They are deep exactly when $-a$ is nonsquare, forming one orbit of size $(q^2 - 1)/2$.

Proof. Assume first that the characteristic is not three. Riemann–Hurwitz gives two totally ramified points t_1, t_2. The confluent Hankel criterion places f in $\mathrm{Osc}(t_1) \cap \mathrm{Osc}(t_2)$; after transporting the pair to $\{0, \infty\}$, the pencil is $\langle T^3, U^3 \rangle$ and a geometric deck generator is $t \mapsto \zeta t$.

For a rational ramification pair, Frobenius fixes the points and the deck map descends exactly when $\zeta^q = \zeta$, namely $q \equiv 1 \pmod 3$. Thus its complement is deep exactly for $q \equiv 2 \pmod 3$. The setwise stabilizer of the ordered normal form is the split dihedral group of order $2(q - 1)$, so the orbit size is

$$\frac{q(q^2 - 1)}{2(q - 1)} = \frac{q(q + 1)}{2}.$$

For a conjugate pair Frobenius interchanges the fixed points and inverts the multiplier; descent is instead $q \equiv 2 \pmod 3$. The nonsplit dihedral stabilizer has order $2(q + 1)$ and the orbit size is $q(q - 1)/2$.

Now suppose the characteristic is three. The second Hasse derivative of

$$\nu(t) = (1, t, t^2, t^3, t^4)$$

is $(0,0,1,3t,6t^2) = e_2$, including in the reversed infinity chart. Thus every osculating plane contains $n = [e_2]$. Substitution shows that n is $\mathrm{PGL}_2(q)$-fixed and $W_n = \langle T^3, U^3\rangle$, so it is deep.

A remaining separable cyclic cover has one wild totally ramified point. Sending it to infinity and comparing coefficients under a nontrivial translation gives

$$W_f = \langle U^3, T^3 + \alpha T^2U + \beta TU^2\rangle, \qquad \alpha = 0, \quad \kappa^2 = -\beta.$$

It is therefore represented by

$$f_a = e_2 - ae_4, \qquad W_{f_a} = \langle U^3, T^3 + aTU^2\rangle.$$

The affine members are $z^3 + az + s$. Such a member has three rational roots exactly when the additive map $z \mapsto z^3 + az$ has a nonzero rational kernel, equivalently when $-a$ is a square. Hence the nonsquare parameters form the deep orbit. Its stabilizer is $t \mapsto \pm t + \beta$, of order $2q$, and orbit–stabilizer gives $(q^2-1)/2$. □

Lemma 4.6 (S_3 stratum)**.** If ϕ_f has geometric monodromy S_3 and $q \geq 23$, then the pencil contains a completely split squarefree cubic.

Proof. By Proposition 4.4, components of Y_f correspond to monodromy orbits on distinct ordered roots. The S_3 action is two-transitive, so Y_f is geometrically integral and has arithmetic genus one. The Aubry–Perret singular-curve bound, in the form

$$\left|\#C(\mathbb{F}_q) - (q+1)\right| \leq 2\pi_C\sqrt{q}$$

for a reduced geometrically integral projective curve of arithmetic genus π_C, gives

$$\#Y_f(\mathbb{F}_q) \geq q + 1 - 2\sqrt{q}$$

[AP95, p. 468]. An integral curve of arithmetic genus one has at most one geometric singular point: if it is singular, its normalization has genus zero and the total delta invariant is one. We therefore delete at most one rational singular point. The diagonal costs

$$Y_f \cdot \Delta = (2,2)\cdot(1,1) = 4.$$

Riemann–Hurwitz gives different degree four for the separable cubic map, hence at most four branch values. A singular cubic has at most two distinct roots and at most two ordered off-diagonal pairs, so the branch deletion costs at most eight. If $q + 1 - 2\sqrt{q} > 13$, a smooth rational point remains outside the diagonal and branch fibers. It gives two distinct rational roots of a squarefree cubic, whose third root is rational. This inequality is equivalent to $q - 2\sqrt{q} > 12$ and holds for every prime power $q \geq 23$. □

Proposition 4.7 (Exhaustion and finite bridge)**.** The preceding cases exhaust every redundancy-five pencil for $q \geq 23$. Certificate R5 closes exactly

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19\}.$$

It finds sporadic S_3-stratum orbits exactly at $q = 7, 8, 9, 11, 13, 17, 19$ and none at $q = 16$.

Proof. The gcd degree is zero, one, or two. Propositions 4.2 and 4.3 settle the positive-gcd cases. In the trivial-gcd case the cubic map is inseparable or separable. The inseparable case occurs only in characteristic three and the overlap equations give the all-cube nucleus pencil. A separable transitive cubic has geometric monodromy C_3 or S_3, settled by Lemmas 4.5 and 4.6. The stated finite residue, canonical representatives, stabilizers, histograms, and Frobenius links are exactly the domain of Certificate R5. □

Table 4: Complete sporadic orbit inventory at redundancy five. A representative $[a_0, \ldots, a_4]$ is in the divided-power syndrome coordinates of Section 3. The last column records the nontrivial Frobenius fusion; a dash means the orbit is fixed.

q	representative	orbit size	stabilizer	Frobenius
7	`[0,1,0,0,3]`	28	12	–
7	`[0,1,0,0,2]`	112	3	–
7	`[0,0,1,0,3]`	168	2	–
7	`[0,1,0,3,2]`	168	2	–
7	`[1,0,0,2,1]`	168	2	–
8	`[0,1,1,0,2]`	252	2	$\to [0,1,1,0,4]$
8	`[0,1,1,0,4]`	252	2	$\to [0,1,1,0,6]$
8	`[0,1,1,0,6]`	252	2	$\to [0,1,1,0,2]$
9	`[1,0,0,1,2]`	180	4	–
9	`[0,1,0,4,1]`	360	2	$\leftrightarrow [0,1,0,4,4]$
9	`[0,1,0,4,4]`	360	2	$\leftrightarrow [0,1,0,4,1]$
11	`[1,0,0,1,10]`	330	4	–
11	`[1,0,0,1,1]`	660	2	–
13	`[0,1,0,0,4]`	182	12	–
13	`[1,0,0,4,7]`	546	4	–
17	`[1,0,0,1,5]`	1224	4	–
19	`[0,1,0,0,4]`	570	12	–

The certificate additionally records the complete cubic-pencil member histogram of each representative. Repeated orbit sizes in Table 4 are therefore visibly distinguished without requiring a working-directory file.

Table 5: Independent R5 completeness totals. The first equality is the direct syndrome enumeration versus the orbit–stabilizer sum; the second decomposes the same total into the formula in Theorem 1.1 and the sporadic rows above.

q	direct = orbit sum	nonsporadic	sporadic
7	889 = 889	245	644
8	1116 = 1116	360	756
9	1391 = 1391	491	900
11	1848 = 1848	858	990
13	2080 = 2080	1352	728
16	2432 = 2432	2432	0
17	4131 = 4131	2907	1224
19	4541 = 4541	3971	570

Corollary 4.8 (The balanced quantum extension at $q = 8$)**.** The 1116 projective deep-hole directions of $\mathrm{PRS}_{\mathbb{F}_8}(4)$ give 1116 relative projective one-column $[10, 5, 6]_8$ MDS extensions. For each extension D,

$$|\Psi_D\rangle = 8^{-5/2} \sum_{c \in D} |c\rangle$$

is a minimum-support AME$(10, 8)$ stabilizer state and determines an associated $[[9, 1, 5]]_8$ quantum MDS code. The directions split as the 360 nonsporadic and 756 sporadic entries of Table 5.

Any product-unitary equivalence between two such equal-phase states is local Clifford, and every transversal conversion between the associated quantum encoders is Clifford factor by factor. In particular, none of the associated quantum MDS codes admits a transversal non-Clifford logical unitary.

Proof. A projective deep-hole direction gives a one-column MDS extension. At $q = 8$, redundancy five turns the $[9, 4, 6]_8$ projective Reed–Solomon code into a $[10, 5, 6]_8$ MDS code, and Table 5 gives the count and its decomposition. The linear MDS–AME construction and the one-party Choi correspondence give the state and the $[[9, 1, 5]]_8$ code [RGRA18]. The local-unitary and transversal claims are the MDS–CSS rigidity and conversion theorems of the companion paper [Rud26d]. □

Remark 4.9. The balance condition for a one-column redundancy-r extension is $q + 2 = 2r$. Among the exact fixed-level classifications at redundancies five through seven, it selects only $(r, q) = (5, 8)$: the formal alternatives $q = 10, 12$ are not prime powers. The projective-semilinear orbit table classifies extensions relative to the distinguished punctured Reed–Solomon code; it is not asserted to be a complete table of local-Clifford or local-unitary state orbits.

5 Coherent polar induction

5.1 Spaces, markers, and bad schemes

Fix a field k, a two-dimensional k-space E, and

$$S_n = \mathbb{P}(\Gamma^n E).$$

For $\lambda \in E^\vee$, define divided-power contraction by

$$\langle \iota_\lambda f, g \rangle = \langle f, \lambda g \rangle.$$

Then

$$g \in W_{\iota_\lambda f} \quad \Longleftrightarrow \quad \lambda g \in W_f. \tag{1}$$

The lower witness lifts squarefreely only when $\lambda \nmid g$. Thus the factor selected for contraction must remain as a marked forbidden root.

Definition 5.1 (Persistent and modular loci)**.** At redundancy j, put $n = j - 1$. The *persistent scheme* $\mathcal{P}_j \subset S_n$ is the determinantal scheme

$$\mathcal{P}_j = \{[f] : \operatorname{rank} C_f^{(2)} \le 2\},$$

cut out by the maximal minors of the three-row catalecticant $C_f^{(2)} : \operatorname{Sym}^{n-2} E^\vee \to \Gamma^2 E$. For every positive-characteristic normal-rational-curve nucleus $\mathcal{N} \subset \mathbb{P}(\Gamma^{n-1} E)$, that is, an intersection of the relevant osculating subspaces, let $\widetilde{\mathcal{N}}$ be its affine cone. Its coherent lift is the linear scheme

$$\mathcal{M}_n(\mathcal{N}) = \mathbb{P} \ker \left(\Gamma^n E \longrightarrow \operatorname{Hom}(E^\vee, \Gamma^{n-1} E / \widetilde{\mathcal{N}}) \right).$$

The *modular locus* $\mathcal{M}_j \subset S_n$ is the reduced union of these coherent lifts; Lucas' binomial criterion determines the nucleus coordinate spans at each level. A *collision divisor* records marker parameters at which the marked factor becomes a repeated root.

Definition 5.2 (Contraction and marker spaces)**.** Let $C_f : E^\vee \to \Gamma^{n-1}E$ be $\lambda \mapsto \iota_\lambda f$, and put $U_f = \mathbb{P}(E^\vee) \setminus \mathbb{P}(\ker C_f)$. The first-polar rational map is

$$\Phi_f : U_f \longrightarrow S_{n-1}, \qquad [\lambda] \longmapsto [\iota_\lambda f].$$

Its image is a point or a line. For $j \geq 1$, let

$$\mathrm{Conf}_j(\mathbb{P}(E^\vee)) = \{(\lambda_1, \dots, \lambda_j) : \lambda_i \neq \lambda_\ell \text{ for } i \neq \ell\}.$$

The ordered marker space $U_{f,j}$ is the open subset on which every successive contraction is nonzero. The universal j-step polar flag is the map

$$(\lambda_1, \dots, \lambda_j) \longmapsto \big([\iota_{\lambda_1} f], \dots, [\iota_{\lambda_j} \cdots \iota_{\lambda_1} f]\big).$$

In an affine coordinate r the first map is

$$\Phi_f(r) = (a_1 - ra_0, \dots, a_n - ra_{n-1}), \qquad \Phi_f(\infty) = (a_0, \dots, a_{n-1}).$$

Example 5.3 (The complete R5-to-R6 propagation at $q = 29$)**.** Work over $\mathbb{F}_{29}$ at redundancy six and take

$$f = (6 : 1 : 0 : 0 : 5 : 1).$$

The Hankel kernel is the net of quartics

$$W_f = \ker \begin{pmatrix} 6 & 1 & 0 & 0 & 5 \\ 1 & 0 & 0 & 5 & 1 \end{pmatrix} = \langle (0,0,1,0,0), (24,1,0,1,0), (28,1,0,0,1) \rangle .$$

The displayed basis has no common factor, so the persistent test does not decide the syndrome. Choose the first marker $r = 0$. Then

$$\iota_0 f = (1 : 0 : 0 : 5 : 1).$$

This is now a redundancy-five syndrome, and its full cubic pencil is

$$W_{\iota_0 f} = \langle (0,1,0,0), (25,0,1,24) \rangle .$$

The member with pencil parameter $[9 : 28]$ is

$$h(t) = 4 + 9t - t^2 + 5t^3 \in W_{\iota_0 f},$$

because the two Hankel products are $4 + 5 \cdot 5 = 0$ and $5(-1) + 5 = 0$ in $\mathbb{F}_{29}$. Direct factorization gives

$$h(t) = 5(t-2)(t-6)(t-27).$$

None of these roots is the marked root 0. Equation (1) therefore lifts h to

$$t\,h(t) = (0,4,9,28,5) \in W_f,$$

a squarefree quartic with roots $0, 2, 6, 27$. The Hankel criterion now places f in the span of the corresponding four normal-rational-curve columns, so f is shallow. This one calculation exhibits every interface in the R5-to-R6 propagation: the upper Hankel net, the ordinary common-factor test, contraction to the complete R5 cubic pencil, selection on its splitting cover, marker avoidance, and squarefree lifting. In the general proof the R5 curve argument supplies such a pencil parameter after deleting the marked fibers; the displayed parameter $[9 : 28]$ is one complete instance of that step.

Definition 5.4 (One-step lower package)**.** Fix $f \in S_n(\mathbb{F}_q)$ with injective contraction map, so that $\Phi_f : \mathbb{P}(E^\vee) \to S_{n-1}$ is defined everywhere. A one-step lower package abstracts the R6 argument: a lower bad locus, the R5 ordered-root curve with its deletions, and the finite set of markers where that construction is unavailable. It consists of:

(i) a closed bad scheme $B_{n-1} \subset S_{n-1}$ and a finite stratification $S_{n-1} \setminus B_{n-1} = \coprod_\alpha X_\alpha$;

(ii) for every $\lambda \in \mathbb{P}(E^\vee)(\mathbb{F}_q)$ with $b = \iota_\lambda f \in X_\alpha(\mathbb{F}_q)$, a specified proper geometrically integral curve $Y_{\alpha,b}$, defined over $\mathbb{F}_q$, with genus bound g_α and point bound

$$\#Y_{\alpha,b}(\mathbb{F}_q) \geq q + \kappa_\alpha - 2g_\alpha\sqrt{q};$$

together with a zero-dimensional closed deletion scheme $D_{\alpha,b,\lambda} \subset Y_{\alpha,b}$, of degree at most δ_α, containing the singular, branch, diagonal, marker, and collision points, such that every rational point outside $D_{\alpha,b,\lambda}$ determines an ordered rational split squarefree $g \in W_b$ with $\lambda \nmid g$;

(iii) a finite parameter scheme $Z_f \subset \mathbb{P}(E^\vee)$, of degree at most d, containing $\Phi_f^{-1}(B_{n-1})$ and every parameter at which the preceding construction is unavailable, including parameters where a fixed factor of the lower system equals a retained marker.

The index α includes any constant-field or Frobenius twist; the assertion in (iii) is the required descent statement. All degrees are geometric degrees on the displayed curve or parameter line.

The construction is summarized in Figure 1. The upper branch is the level-specific component calculation; the lower branch is the general transverse argument.

$$\begin{array}{ccccc}
[f] \in S_n & \dashrightarrow & \Phi_f(U_f) \subset S_{n-1} & \longrightarrow & \text{lower ordered splitting cover} \\
 & & \searrow \text{ contained} & & \text{persistent/Lucas component,} \\
 & & \nearrow \text{ transverse} & & \text{delete branch, diagonal, and markers} \\
 & & & & \downarrow \text{ lift by (1)} \\
 & & & & \text{split squarefree witness in } W_f.
\end{array}$$

Figure 1: Coherent polar induction. The contained branch is a level-specific component theorem; only the transverse branch is controlled by the general point-count argument.

Theorem 5.5 (Polar-flag construction)**.** The maps of Definition 5.2 commute with field extension and the natural $\mathrm{GL}(E)$ action. They include the infinity chart displayed above. If $g \in W_{\iota_{\lambda_j}\cdots\iota_{\lambda_1}f}$, then $\lambda_1 \cdots \lambda_j g \in W_f$; the lift is squarefree exactly when g is squarefree and avoids every marker λ_i.

Proof. Contraction is defined by the perfect divided-power/symmetric-power pairing, so it commutes with base change and change of basis. Repeated application of (1) gives the membership assertion. Since the λ_i are distinct on the configuration space, their product is squarefree; the product with g is squarefree precisely when g is squarefree and none of the λ_i divides it. The coordinate formula at infinity follows by evaluating C_f on the second basis covector. □

Lemma 5.6 (Three readings of the collision divisor)**.** Assume W_f has trivial gcd, and let $\lambda \in \mathbb{P}(E^\vee)$. The following are equivalent:

(i) every member of W_f divisible by λ is divisible by λ^2;

(ii) λ is a fixed factor of $W_{\iota_\lambda f}$;

(iii) λ lies on the collision divisor of the linear series W_f.

Proof. Equation (1) gives the equality of subspaces

$$\lambda W_{\iota_\lambda f} = W_f \cap \lambda \operatorname{Sym}^{n-2} E^\vee.$$

Thus (i) and (ii) are the same statement after dividing by λ. Since W_f is base-point-free, (i) says exactly that the morphism defined by W_f has zero differential at λ, which is (iii). $\square$

Lemma 5.7 (Uniform collision alternatives)**.** Let $j \geq 5$, and let a base-point-free g^{j-4}_{j-2} on $\mathbb{P}^1$ define an inseparable morphism in characteristic p. Then

$$p \mid (j-2), \qquad p(j-4) \leq j-2.$$

The only possible inseparable cases are $(j,p) = (5,3)$ and $(6,2)$. In the separable case the collision divisor has degree at most $2j-6$; in particular this holds unconditionally for $j \geq 7$.

Proof. An inseparable morphism factors through absolute Frobenius. Hence $p \mid (j-2)$, and its $j-3$ sections lie in the $(j-2)/p+1$-dimensional space of p-th powers. This gives the inequality. Since $(j-2)/(j-4) < 2$ for $j \geq 7$, no prime remains; the two smaller cases follow directly. In the separable case some value–derivative minor is nonzero; as a binary Wronskian it has degree $2(j-2)-2 = 2j-6$ and contains the collision scheme. $\square$

Lemma 5.8 (Linear modular pullbacks)**.** Let $\mathcal{N} \subset S_{j-2}$ be a projective linear modular nucleus. Then $\Phi_f^{-1}(\mathcal{N})$ is empty, one point, or the whole parameter line. In the transverse case its degree is at most one.

Proof. The polar map has linear coordinates in the parameter, so the pullback ideal is generated by linear forms on $\mathbb{P}^1$. $\square$

Lemma 5.9 (Old-marker fixed-factor divisor)**.** Assume W_f has trivial gcd, and fix a retained marker $x \in \mathbb{P}(E^\vee)$. The parameters λ for which x divides every member of $W_{\iota_\lambda f}$ form a proper closed subscheme of $\mathbb{P}(E^\vee)$ of degree at most three.

Proof. Append the evaluation row at x to the two consecutive Hankel rows of $\iota_\lambda f$. The maximal minors cut out the stated fixed-factor condition. Their entries are linear in λ, so their common zero scheme has degree at most three.

The minors cannot vanish identically. By Equation (1), identical vanishing would say

$$W_f \cap \lambda \operatorname{Sym}^{n-2} E^\vee \subset x \operatorname{Sym}^{n-2} E^\vee \qquad \text{for every geometric } \lambda.$$

Given a nonzero $g \in W_f$, choose a geometric linear factor $\lambda \mid g$. The displayed containment then gives $x \mid g$. Thus x would divide every member of W_f, contrary to the trivial-gcd hypothesis. $\square$

Lemma 5.10 (Exact-linear-gcd transport)**.** Suppose $W_f = LV$ has exact linear gcd L, and let $\lambda \neq L$. Then every member of $W_{\iota_\lambda f}$ is divisible by L. Along a coherent flag chosen outside the rank-at-most-two terminal carrier, the gcd remains exactly L; thus the flag reaches the R5 quadratic-graph package with the same fixed factor.

Proof. The lifting identity gives

$$\lambda W_{\iota_\lambda f} = W_f \cap \lambda \operatorname{Sym}^{n-2} E^\vee \subset L \operatorname{Sym}^{n-2} E^\vee.$$

Since L and λ are coprime, division by λ shows that L divides every lower member. If the lower gcd grows to degree two, the lower syndrome lies in the rank-at-most-two terminal carrier and that parameter is excluded. The dimension bound rules out a larger common factor away from the rank-one boundary. Induction over the marker list proves the assertion. Excluding $\lambda = L$ is exactly the fixed-factor collision needed to keep the final lifted product squarefree. □

Lemma 5.11 (Identically colliding stages)**.** If the collision divisor of a base-point-free stage is the whole parameter line, then the stage is inseparable. The only possibilities are the characteristic-three R5 component and the characteristic-two R6 component, and their coherent lifts lie in the declared persistent or modular locus.

Proof. Lemma 5.6 identifies whole-line collision with zero differential of the moving linear series. The numerical argument in Lemma 5.7 leaves only $(j,p) = (5,3)$ and $(6,2)$. Proposition 4.7 identifies the first with the wild rank/fixed-factor boundary, while Propositions 6.4 and 6.5 identify the second with the binary modular component. Proposition 5.21 and Lemma 5.8 give their coherent lifts. □

Definition 5.12 (Contained pullback and recursive bad scheme)**.** Let $B \subset S_{n-1}$ be a closed scheme with homogeneous ideal I_B. On the open $S_n^\circ \subset S_n$ where C_f is injective, pull each element of I_B back along the universal polar map

$$\mathbb{P}(E^\vee) \times S_n^\circ \longrightarrow S_{n-1}.$$

Expand the resulting bihomogeneous forms in the marker coordinates. The vanishing of all their coefficients defines the *contained-pullback scheme* $\operatorname{Cont}_n(B)$: its geometric points are exactly the f for which the whole polar line lies scheme-theoretically in B. This coefficient ideal is independent of the chosen generators of I_B.

Let $\mathcal{B}_5^{\mathrm{red}} \subset S_4$ be the reduced terminal carrier consisting of the rank-at-most-two Hankel locus, the closure of the cyclic carrier on the rank-two open, its true characteristic-two cyclic plane and characteristic-three wild cone, the rank-one contraction boundary, and the two identically colliding inseparable components. The exact-linear-gcd stratum is not part of this carrier: Proposition 4.3 supplies its separate quadratic-graph lower package. Ordinary collision points are deletion divisors, not components of $\mathcal{B}_5^{\mathrm{red}}$.

Inductively, let $\mathcal{B}_{j+1}^{\mathrm{red}} \subset S_j$ be the reduced union of the closure of $\operatorname{Cont}_j(\mathcal{B}_j^{\mathrm{red}})$, the noninjective-contraction locus, and the locus on which the universal marker-collision divisor is the whole parameter line. We call this the *reduced recursively pointed contained bad locus* at redundancy $j+1$.

Definition 5.13 (One-step contained-component assertion)**.** For a specified bad scheme B_{n-1}, collision divisor, and explicit closed list $\mathcal{C}_n \subset S_n$, the assertion $\mathrm{CC}(n,1;\mathcal{C}_n)$ says that every f for which the contraction map is noninjective and hence supplies no polar line, $\Phi_f^* B_{n-1}$ is the whole parameter line and hence supplies no transverse marker, or the collision divisor is the whole line and hence every lift repeats its marked root, belongs to $\mathcal{C}_n$. When the list is clear we write $\mathrm{CC}(n,1)$.

Definition 5.14 (Stable-component assertion)**.** The assertion $\mathrm{SC}(j)$ says that every irreducible component of $\mathcal{B}_j^{\mathrm{red}}$ is contained in $\mathcal{P}_j \cup \mathcal{M}_j$. It is a geometric irreducible-component statement and contains no field-size threshold; it makes no claim about embedded primary structure in a nonreduced recursive thickening.

Proposition 5.15 (Exact bottom carrier ledger)**.** The reduced terminal carrier $\mathcal{B}_5^{\mathrm{red}}$ is exactly the locus on which neither the quadratic-graph package of Proposition 4.3 nor the geometrically integral off-diagonal package of Lemma 4.6 applies. Over an algebraic closure, each of its reduced irreducible components is one of the following:

(i) the rank-at-most-two Hankel component or its rank-one boundary;

(ii) the noncollision cyclic carrier;

(iii) in characteristic two, the true cyclic plane or one of the two ordered-Hessian rulings;

(iv) in characteristic three, the wild cone or its ruling boundary;

(v) an identically colliding inseparable component.

There is no exact-linear-gcd component in this list.

Proof. Separate the cubic pencils by their gcd degree. Degree at least two is the rank-at-most-two Hankel component by Proposition 4.2; degree one has the quadratic-graph package of Proposition 4.3. On the trivial-gcd locus, Proposition 4.4 identifies failure of the geometrically integral off-diagonal package with reducible geometric monodromy or inseparability.

For reducible monodromy, divide the alternating evaluation determinant by the diagonal bracket. Its moving part has bidegree $(2, 2)$. A reducible moving part has a vertical or horizontal factor, which returns to the excluded fixed-factor package, or two $(1, 1)$ factors. Symmetry under exchange of the roots makes the factors individually symmetric, exchanged, or anti-invariant. The symmetric case has a rank-one factor; the exchanged case has either a rank-one factor or the cyclic equation; and the anti-invariant case survives the Plücker relation only in characteristic three, where it is diagonal collision. These are the three factorizations displayed in (5) and (6) below. Thus no further characteristic-zero component occurs.

Primitive specialization of the cyclic equations has only the two vertical primes described by the ordered-Hessian calculation in characteristic two; they are the true cyclic plane and the two stated rulings. In characteristic three the inseparable closure is the wild cone. Lemma 5.11 gives the remaining whole-line collision components. The integral bridge (10), its saturation (11), and the vertical calculations below verify these reduced component statements scheme-theoretically, rather than only on geometric points. $\square$

Table 6 records the disposition used in the all-level proof. The last two rows are transverse packages and are not components of $\mathcal{B}_5^{\mathrm{red}}$.

Table 6: Bottom carriers and their all-level disposition.

Bottom locus	Exact test	Disposition after coherent lifting
Rank at most two	$D = 0$; Hankel minors	Persistent rank-two locus by Proposition 5.21
Tame cyclic carrier	$V_1 = \cdots = V_7 = 0$, $D \neq 0$	Persistent locus or rank one by (12)
Binary cyclic plane	$c_0 = c_4 = 0$	Lucas modular kernel; empty after redundancy seven for this descendant
Binary Hessian rulings	the two displayed quadratic rulings	Persistent Veronese ruling or rank one by consecutive overlap
Ternary wild cone	$\mathrm{Join}(e_2, C_4)$	Rank one by the ruling calculation
Whole-line collision	zero differential	The two R5/R6 persistent or modular components of Lemma 5.11
Exact linear gcd	common factor L	Factor-preserving flag and R5 quadratic graph; not a terminal carrier
Ordinary collision	nonzero Wronskian divisor	Finite parameter deletion; not a terminal carrier

The indices follow two conventions. The assertion $\mathrm{CC}(n, 1; \mathcal{C}_n)$ is indexed by the divided-power degree n, so $f \in S_n$ has redundancy $n+1$; $\mathrm{SC}(j)$ is indexed directly by redundancy. Accordingly, after the two base cases below, $\mathrm{SC}(j)$ is the recursive content of $\mathrm{CC}(j-1, 1; \mathcal{P}_j \cup \mathcal{M}_j)$.

Lemma 5.16 (Recursive transport to the bottom ledger)**.** Let $j \geq 6$, and let f lie on an irreducible component of $\mathcal{B}_j^{\mathrm{red}}$. Either f lies in a noninjective, identically colliding, persistent, or modular component already declared at an intermediate level, or every ordered sequence of $j-5$ distinct markers for which the contractions are defined has its bottom syndrome in $\mathcal{B}_5^{\mathrm{red}}$. In the latter case,

$$\mathbb{P}(\operatorname{rowspace} \operatorname{Cat}_{j-5,4}(f)) \subset \mathcal{B}_5^{\mathrm{red}}.$$

Proof. Proceed down the recursive definition of $\mathcal{B}_j^{\mathrm{red}}$. At one level an irreducible component lies in one member of the finite reduced union defining the bad locus. The noninjective member gives the first alternative, and Lemma 5.11 classifies the whole-line collision member. Otherwise the component lies in a contained pullback, so its whole polar line maps into the preceding bad locus. The image of that line is irreducible and hence lies in one irreducible component of the preceding finite reduced union. If that component is persistent or modular, Proposition 5.21 or Lemma 5.8 lifts it to the corresponding declared component upstairs. Exact linear gcd does not occur in the recursive carrier: it is a transverse terminal stratum with its own graph package. Ordinary collision is a finite deletion divisor, while whole-line collision was already classified. In every other case the image lies in the next contained pullback. Repeating this finite descent reaches $\mathcal{B}_5^{\mathrm{red}}$, and proves the assertion for every ordered tuple of distinct markers.

Products of $j-5$ distinct linear forms are dense in $\mathbb{P}(\operatorname{Sym}^{j-5} E^\vee)$. The bottom contraction is the linear map whose matrix is $\operatorname{Cat}_{j-5,4}(f)$. Taking its scheme-theoretic image closure therefore gives the displayed rowspace containment. The projective kernel removes only a proper closed subset of the source on each nonzero branch; if the map is zero, f is already in the noninjective boundary. $\square$

Proof of Theorem 1.3. Fix an irreducible component and work at its geometric generic point f. Lemma 5.16 handles any noninjective, collision, persistent, or modular branch encountered during the descent. We may therefore assume its second alternative. Put $m = j-5$ and write

$$\operatorname{Cat}_{m,4}(f) = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_m & a_{m+1} & a_{m+2} & a_{m+3} & a_{m+4} \end{pmatrix}.$$

If $h = (h_0, \ldots, h_m)$ is the coefficient vector of the marker product, its bottom syndrome is

$$c(h) = h \operatorname{Cat}_{m,4}(f). \tag{2}$$

Over an algebraic closure, products of m distinct linear forms are dense in $\mathbb{P}(\operatorname{Sym}^m E^\vee)$. The map in (2) is linear and homogeneous. After deleting its projective kernel, the scheme-theoretic closure of the attainable bottom syndromes is therefore

$$\overline{\{c(h) : h \text{ is a product of distinct markers}\}} = \mathbb{P}(\operatorname{rowspace} \operatorname{Cat}_{m,4}(f)). \tag{3}$$

This includes the homogeneous boundary at infinity. Marker diagonals are removed only from the dense source open and reappear in its closure. The recursive-transport lemma places the right-hand side in the finite bottom bad union. Since that projective row space is irreducible, it lies in one irreducible component of the union. We now construct and check that component ledger.

Integral bottom carrier. Let $\mathbf{G} = \mathrm{Gr}(2, \Gamma^3 E)$ over $\operatorname{Spec} \mathbf{Z}$, and write

$$(z_0, \ldots, z_5) = (p_{01}, p_{02}, p_{03}, p_{12}, p_{13}, p_{23}), \qquad z_0z_5 - z_1z_4 + z_2z_3 = 0.$$

After dividing the alternating evaluation determinant of a cubic pencil by the diagonal bracket, its ordered-root incidence is

$$\begin{aligned} G_z(t,s) = z_0 + z_1(t+s) + z_2(t^2 + ts + s^2) + z_3 ts \\ + z_4 ts(t+s) + z_5 t^2 s^2. \end{aligned} \tag{4}$$

Its homogenization is intrinsic on $\mathbb{P}(E^\vee) \times \mathbb{P}(E^\vee)$. We retain the closed collision incidence before taking a residual.

Over an algebraically closed field, a reducible moving part either has a vertical or horizontal factor, hence a fixed-root graph stratum or collision deletion, or splits into two $(1,1)$ factors. Symmetry under $\tau(t,s) = (s,t)$ gives three cases. If both factors are symmetric,

$$L = a + b(t+s) + cts, \qquad M = A + B(t+s) + Cts,$$

then

$$z_0z_5 - z_1z_4 + z_2z_3 = (ac - b^2)(AC - B^2), \tag{5}$$

so one factor has rank one and gives collision. If the factors are exchanged,

$$L = a + bt + cs + dts, \qquad G_z = L(t,s)L(s,t),$$

then

$$z_0z_5 - z_1z_4 + z_2z_3 = (ad - bc)(ad - b^2 + bc - c^2). \tag{6}$$

The first factor is again collision; the second is the cyclic branch. Finally, if $\tau L = -L$, then L is proportional to the homogeneous diagonal bracket. The product of the two anti-invariant factors has

$$(z_0, \ldots, z_5) = (0, 0, -\mu, 3\mu, 0, 0)$$

and Plücker value $-3\mu^2$. It survives only in characteristic three, where it belongs to the retained diagonal collision incidence. These three cases are exhaustive by unique factorization.

For a bottom syndrome $c = (c_0, \ldots, c_4)$, the kernel line of

$$H_c = \begin{pmatrix} c_0 & c_1 & c_2 & c_3 \\ c_1 & c_2 & c_3 & c_4 \end{pmatrix}$$

has Plücker coordinates

$$\begin{aligned} z_0 &= c_2c_4 - c_3^2, & z_1 &= -(c_1c_4 - c_2c_3), & z_2 &= c_1c_3 - c_2^2, \\ z_3 &= c_0c_4 - c_1c_3, & z_4 &= -(c_0c_3 - c_1c_2), & z_5 &= c_0c_2 - c_1^2. \end{aligned} \tag{7}$$

Eliminating a, b, c, d from the noncollision factor of (6) gives

$$\begin{aligned} K_1 &= 3z_4^2 - 9z_2z_5 - z_3z_5, & K_2 &= z_3z_4 - 3z_1z_5, \\ K_3 &= z_3^2 - 9z_0z_5, & K_4 &= z_2z_3 - z_1z_4 + z_0z_5, \\ K_5 &= z_1z_3 - 3z_0z_4, & K_6 &= 3z_1^2 - 9z_0z_2 - z_0z_3. \end{aligned} \tag{8}$$

Thus $J = (K_1(z(c)), \ldots, K_6(z(c)))$. The primitive cyclic ideal $V = (V_1, \ldots, V_7)$ is

$$\begin{aligned} V_1 &= 2c_3^3 - 3c_2c_3c_4 + c_1c_4^2, \\ V_2 &= 6c_2c_3^2 - 9c_2^2c_4 + 2c_1c_3c_4 + c_0c_4^2, \\ V_3 &= 2c_1c_3^2 - 3c_1c_2c_4 + c_0c_3c_4, \\ V_4 &= c_0c_3^2 - c_1^2c_4, \\ V_5 &= 2c_1^2c_3 - 3c_0c_2c_3 + c_0c_1c_4, \\ V_6 &= 6c_1^2c_2 - 9c_0c_2^2 + 2c_0c_1c_3 + c_0^2c_4, \\ V_7 &= 2c_1^3 - 3c_0c_1c_2 + c_0^2c_3. \end{aligned} \tag{9}$$

Put

$$D = \det \begin{pmatrix} c_0 & c_1 & c_2 \\ c_1 & c_2 & c_3 \\ c_2 & c_3 & c_4 \end{pmatrix}.$$

Direct integral expansion gives

$$J \subset V, \qquad 3DV \subset J. \tag{10}$$

The supplement prints the 6×7 coefficient matrix for the second containment. Integer reduction also gives $DV \not\subset J$, and the $\mathbf{F}_3$-point $(1, 0, 0, 1, 0)$ lies on J, has $D = 2$, and does not lie on V. Hence 3 is minimal for this elimination presentation, not asserted to be an intrinsic invariant.

Over $\mathbf{Z}[1/6]$, diagonal rescaling identifies (9) with the prime homogeneous kernel of

$$[u : v : w] \longmapsto [u^2 : 2uv : v^2 + 2uw : 2vw : w^2].$$

The image point $(1, 0, 2, 0, 1)$ has $D = -6$, so $D \notin V$ and $V : D^\infty = V$. Therefore

$$(J : D^\infty) = (V : D^\infty) \text{ over } \mathbf{Z}[1/3], \qquad (J : D^\infty) = V \text{ over } \mathbf{Z}[1/6]. \tag{11}$$

The inversion of 2 in the second statement is necessary: in characteristic two, V retains a D-supported component, so $(V : D^\infty) \neq V$.

Coherent lifts and exceptional fibers. For a homogeneous carrier ideal I, define its *coherent-Fano ideal* by substituting the universal consecutive polar line into every generator of I, expanding in the two line parameters, and setting all coefficients to zero. It represents upper syndromes whose entire polar line lies in the carrier, with the required consecutive-Hankel integrability constraint.

For $d = (d_0, \dots, d_5)$, substitute $c_i = xd_i + yd_{i+1}$ in (9), and let $F_{i,k}$ be the coefficient of $x^{3-k}y^k$ in $V_i(c(x, y))$. If $H_1, \dots, H_4$ are the maximal minors, in column order, of the consecutive 3×4 Hankel matrix of d, then

$$\begin{aligned} 6H_1 &= -3F_{4,0} - 2F_{5,1} + F_{6,2} - 6F_{7,3}, \\ 6H_2 &= -6F_{3,0} + 9F_{4,1} + 6F_{5,2} - 3F_{6,3}, \\ 6H_3 &= -3F_{2,0} + 6F_{3,1} - 9F_{4,2} - 6F_{5,3}, \\ 6H_4 &= -6F_{1,0} + F_{2,1} - 2F_{3,2} + 3F_{4,3}. \end{aligned} \tag{12}$$

The sign pattern is checked by reversal: $H_1 \leftrightarrow H_4$, $H_2 \leftrightarrow H_3$, $V_1 \leftrightarrow V_7$, $V_2 \leftrightarrow V_6$, $V_3 \leftrightarrow V_5$, while $V_4 \mapsto -V_4$. Thus magnitudes reverse and exactly the F_4 position changes sign in each pair. Equation (12) puts the tame cyclic coherent-Fano locus inside the persistent locus over $\mathbf{Z}[1/6]$.

We next treat the vertical fibres. In characteristic three the true wild carrier is the cone $\mathrm{Join}(e_2, C_4)$. Any line not through its vertex would project to a line in the quartic normal rational curve, so every contained line is a ruling. Move a ruling to $\langle e_2, e_4 \rangle$. The conditions

$$(a_0, \dots, a_4), (a_1, \dots, a_5) \in \langle e_2, e_4 \rangle$$

give $a_0 = a_1 = a_3 = 0$ and $a_1 = a_2 = a_4 = 0$, leaving only a_5. Undoing the frame gives $f = \mu\nu_5(t)$, including infinity, hence the rank-one/fixed-factor boundary.

In characteristic two the true cyclic component is $\Pi_{\mathrm{cyc}} = (c_0, c_4)$; the other component is supported on $D = 0$. The remaining inseparable degeneration is seen on the *ordered-Hessian chart*: for an ordered pencil basis p_0, p_1, it is the bidegree-$(2, 2)$ equation

$$u^2 \operatorname{Hess}^{\mathrm{div}}(p_0) + uv \operatorname{PolHess}^{\mathrm{div}}(p_0, p_1) + v^2 \operatorname{Hess}^{\mathrm{div}}(p_1) = 0.$$

The tangent quadric has two conic families of generator lines, called the tangent-quadric rulings:

$$z_1 = z_4 = 0, \quad z_2 = z_3, \quad z_0 z_5 = z_2^2,$$
$$z_0 = z_5 = 0, \quad z_2 = z_3, \quad z_1 z_4 = z_2^2.$$

The first is the square-quadratic boundary of the persistent Veronese. For the complementary ruling, the two consecutive identities

$$h_{-1}^2 = h_{-2}h_0, \qquad h_1^2 = h_0 h_2$$

propagate across their common term and make all five contraction forms proportional, hence rank one. Vertical and horizontal factors are, respectively, ordinary collision and exact-linear-gcd strata. They are outside the terminal carrier and are handled by the deletion divisor and quadratic-graph package.

Componentwise rowspace conclusion. It remains to apply the rowspace reduction (3). If the row space lies in the rank-two Hankel component, induct on m: every first contraction has its $(m-1)$-marker row space in the same component, and Proposition 5.21 then puts f in the upper persistent scheme. For a linear modular component $L_S = \langle e_i : i \in S\rangle \subset \Gamma^d E$, the two contractions of an upper basis vector e_q are, up to divided-power normalization, e_q and e_{q-1}. Consequently its coherent lift is the scheme-theoretic kernel

$$\ker\Big(\Gamma^{d+1}E \longrightarrow \mathrm{Hom}(E^\vee, \Gamma^d E/L_S)\Big) = \langle e_q : q, q-1 \in S\rangle.$$

After ℓ lifts its support is therefore exactly the set of q for which $q, q-1, \ldots, q-\ell \in S$. For an NRC nucleus, Lucas' theorem computes S from the vanishing binomial digits. Thus the coherent lifts are precisely the Lucas consecutive-support components: the kernel description is scheme-theoretic, commutes with base change, and leaves neither an extra nonlinear component nor an embedded one. The tame projected Veronese contains no line, so its rowspace pullback has rank one. A row space in the characteristic-three cone is a ruling or a point; applying the displayed two-row overlap to each adjacent pair again gives rank one.

Finally, in characteristic two, rowspace containment in Π_{cyc} forces the first and last columns of the catalecticant to vanish:

$$a_0 = \cdots = a_m = 0, \qquad a_4 = \cdots = a_{m+4} = 0.$$

For $m = 1, 2$ these are the known N_3 and $[e_3]$ pullbacks. For $m \geq 3$ the blocks cover every coefficient, so no projective point remains. Hence the characteristic-two modular component descended from the R5 cyclic plane terminates after redundancy seven. This does not erase the fresh higher Lucas carriers of the separate Lucas-carrier analysis; for example $\mathbb{P}\langle e_2, \ldots, e_7\rangle \subset \Gamma^9 E$ is nonempty.

Whole-line collision is covered by Lemma 5.11; ordinary collision and exact linear gcd are the two transverse terminal packages excluded from $\mathcal{B}_5^{\mathrm{red}}$. This exhausts the finite bottom component ledger and proves $\mathrm{SC}(j)$ for every $j \geq 6$, together with the asserted termination of the cyclic-plane descendant. $\square$

The rank-one boundary, ordinary rank-two component, collision alternative, and modular pullback are uniform by Proposition 5.21 and Lemmas 5.7 and 5.8. The two inseparable collision possibilities are exactly the characteristic-three R5 case and the characteristic-two R6 case treated in Propositions 4.7 and 6.4. Section 6 independently recovers the redundancies-six and seven base cases.

Theorem 5.17 (One-step polar escape)**.** Let $f \in S_n(\mathbb{F}_q)$ have a one-step lower package as in Definition 5.4. Put

$$\mathcal{H}_\kappa(g,\delta) = 1 + \left\lfloor \left(g + \sqrt{g^2 + \delta - \kappa}\right)^2 \right\rfloor$$

and assume $\delta_\alpha \geq \kappa_\alpha - g_\alpha^2$ for every α. If

$$q \geq \max_\alpha \mathcal{H}_{\kappa_\alpha}(g_\alpha, \delta_\alpha) \quad \text{and} \quad q + 1 > d,$$

then W_f contains a split squarefree member.

Proof. Since $\#\mathbb{P}^1(\mathbb{F}_q) = q + 1 > d$, choose $\lambda \notin Z_f(\mathbb{F}_q)$. On the corresponding lower stratum the recorded point bound gives

$$q + \kappa_\alpha - 2g_\alpha\sqrt{q} > \delta_\alpha,$$

so there is a rational point outside all deletions. It yields a split squarefree lower member avoiding λ. Equation (1) lifts it to a split squarefree member upstairs, so the original system is shallow. $\square$

The theorem isolates the numerical escape and one-step lifting argument. Iterating it requires the contained-component assertion at every intermediate level, not only at the top.

Recursively, the current contraction is chosen outside $\mathcal{P}_j \cup \mathcal{M}_j$; the uniform alternatives and the next SC theorem make its polar line transverse, so a new marker exists off the finite pullback. No inverse-image stability of the declared loci is assumed.

Proposition 5.18 (Uniform iterated lower packages)**.** Fix $r \geq 6$, and start with a redundancy-r syndrome outside $\mathcal{P}_r \cup \mathcal{M}_r$. At every stage $6 \leq j \leq r$ of a transverse polar flag, maintain the invariant that the current syndrome lies outside $\mathcal{P}_j \cup \mathcal{M}_j$ by including their lower pullbacks in the unavailable-parameter scheme. Definition 5.4 then applies with a proper zero-dimensional parameter scheme. If $s = r - j$ markers have already been retained, its degree satisfies

$$d_6 \leq 3 + 4 + 6 + 3(r - 6) = 3r - 5$$

at the bottom stage and

$$d_j \leq 3 + 1 + (2j - 6) + 3(r - j) = 3r - j - 2 \quad (j \geq 7).$$

If an exact linear gcd L appears at stage j, the separate factor-preserving flag of Lemma 5.10 has parameter degree

$$e_j \leq 3 + 1 + (r - j) = r - j + 4:$$

the terminal-carrier pullback, the point L, and the old markers. The final R5 ordered-root curve is proper and geometrically integral, has $(g, \kappa) = (1, 1)$, and its deletion scheme has degree at most

$$13 + 6(r - 5) = 6r - 17.$$

If the terminal pencil instead has exact linear gcd, its quadratic deck graph is a proper geometrically integral rational curve with exactly $q+1$ rational points, and its deletion scheme has degree at most

$$8 + 2(r - 5) = 2r - 2.$$

Proof. Suppose first that a stagewise pullback or collision scheme were the whole parameter line. The recursively pointed definition and the stable-component calculation above, together with Proposition 5.21 and Lemmas 5.11 and 5.8, would place the current syndrome in its persistent or modular component, contrary to the maintained invariant. Choosing the next marker outside the finite pullbacks preserves that invariant; membership of one selected contraction is never lifted backwards. Every unavailable-parameter scheme is therefore a proper closed subscheme of $\mathbb{P}^1$, hence is zero-dimensional.

The ordinary terminal-carrier pullback has degree at most three. Lemma 5.9 shows separately that each old marker contributes a proper divisor of degree at most three. At $j = 6$, the cyclic/wild pullback has degree at most four and the collision divisor degree at most six by Propositions 6.4 and 6.5. This gives the first displayed bound. For $j \geq 7$, the modular pullback has degree at most one by Lemma 5.8, and the collision divisor has degree at most $2j - 6$ by Lemma 5.7; this gives the second bound.

On the exact-linear-gcd branch, Lemma 5.10 preserves the factor L. The rank-at-most-two pullback costs at most three; if it were the whole line, Proposition 5.21 would put the upper syndrome in the excluded persistent locus. Deleting L and the $r - j$ old markers gives $e_j \leq r - j + 4$. Thus this branch also reaches the R5 pencil without entering a claimed contained component.

Finally, Proposition 4.4 and Lemma 4.6 give the proper geometrically integral genus-one off-diagonal curve. Its singular, diagonal, and branch deletions have total degree thirteen. Vanishing at one retained marker cuts at most six ordered pairs, so the $r - 5$ markers add at most $6(r-5)$. These schemes include every singular, branch, diagonal, marker, and fixed-factor collision excluded in Definition 5.4.

If the terminal pencil has exact linear gcd, the separable quadratic map and its rational deck involution give the proper integral graph in Proposition 4.3. Its two fixed points, four branch-fiber points, and two incidences with the gcd root cost at most eight. Avoiding each retained marker removes at most the two graph points in which it occurs. Thus the graph deletion degree is at most $8+2(r-5)$. The inseparable quadratic case is rank one by the overlapping Hankel calculation in that proposition and has already been routed to the noninjective boundary. ☐

Remark 5.19 (Point-count convention). The correction κ records the exact lower bound available for the chosen model. Every genus-one curve retained in this paper uses the Aubry–Perret bound [AP95, p. 468] with $\kappa = 1$; a possible rational singular point is included instead in the deletion degree.

Theorem 5.20 (Uniform transverse threshold)**.** Fix a redundancy $r \geq 6$. If

$$q \geq Q_r := 6r - 15 + \left\lfloor 2\sqrt{6r-17} \right\rfloor,$$

every split-free redundancy-r syndrome belongs to $\mathcal{P}_r \cup \mathcal{M}_r$.

Proof. Proposition 5.18 supplies the one-step lower package at every stage of the pointed contraction to the R5 cubic pencil. At the bottom its deletion bound is

$$\delta_r = 13 + 6(r-5) = 6r - 17, \qquad \mathcal{H}_1(1, \delta_r) = 6r - 15 + \left\lfloor 2\sqrt{6r-17} \right\rfloor.$$

Its stagewise degree bounds give

$$d_6 \leq 3 + 4 + 6 + 3(r-6) = 3r - 5.$$

For $j \geq 7$, they give

$$d_j \leq 3 + 1 + (2j-6) + 3(r-j) = 3r - j - 2 < 3r - 5.$$

Thus the maximum is attained at the bottom stage, and $Q_r > 3r - 5$ makes the parameter-line inequality $q + 1 > d_j$ automatic. The exact-linear-gcd flag has $e_j \le r - j + 4 \le r - 2 < 3r - 5$, so its parameter line is also nonbinding. The terminal graph on this branch has, by Proposition 5.18,

$$\delta_r^{\mathrm{lin}} = 8 + 2(r-5) = 2r - 2, \qquad Q_r + 1 - \delta_r^{\mathrm{lin}} = 4r - 12 + \lfloor 2\sqrt{6r - 17} \rfloor > 0.$$

Thus this comparison is weaker for every $r \ge 6$.

Apply Theorem 5.17 successively. If a recursively pointed bad scheme contains the whole polar line, the uniform alternatives above and $\mathrm{SC}(j)$ place the syndrome in the declared persistent or modular list. Otherwise a marker remains outside its finite scheme. After reaching the R5 pencil, either its exact-linear-gcd graph or its trivial-gcd off-diagonal curve supplies a split squarefree cubic avoiding every marker by the two displayed point bounds. Theorem 5.5 lifts it to the original system. □

Uniform radius gate. The coding interpretation of this range is unconditional. Apply Seroussi–Roth [SR86, Thm. 1] to the dual r-dimensional GRS code. It gives nonextendability when

$$r \le \left\lfloor \frac{q}{2} \right\rfloor + 2,$$

apart from the even-field dimension-three exception. For $r \ge 6$,

$$Q_r \ge 6r - 15 \ge 2r - 3,$$

so $q \ge Q_r$ gives $2r \le q + 3 \le q + 4$, equivalently the displayed dimension inequality, and cannot meet the exception. Dür's completeness–covering-radius equivalence [Dür94], in the form stated by Kaipa [Kai17, Sec. IV and Prop. 4(1)], then gives $\rho(\mathrm{PRS}(q + 1 - r)) = r - 1$. Corollary 3.2 therefore identifies the split-free directions in Theorem 5.20 with the projective deep-hole directions.

For $r = 5, 6, 7$, the integer cutoffs supplied by the same formula are

$$Q_5 = 22, \qquad Q_6 = 29, \qquad Q_7 = 37;$$

the first prime powers meeting them are $23, 29, 37$. Thus the former threshold table consisted of three evaluations of one formula with $Q_r = 6r + O(\sqrt{r})$. The same calculation recovers the verified R6 and R7 parameter bounds 13 and 16. In general the parameter line is nonbinding; the stable-component assertion is supplied uniformly by Theorem 1.3. Propositions 6.7 and Corollary 6.14 discharge the required assertions at redundancies six and seven.

5.2 Contained-component calculations

The ordinary persistent locus admits a uniform contained-line calculation. This does not classify other lower bad schemes, but it closes the persistent part of every level-specific component gate.

Proposition 5.21 (Contained rank-two polar lines)**.** Let $n \ge 5$, let $f \in \mathbb{P}(\Gamma^n E)$, and let

$$A = C_f^{(2)} : \mathrm{Sym}^{n-2} E^\vee \longrightarrow \Gamma^2 E$$

be the three-row catalecticant. Assume the contraction map $E^\vee \to \Gamma^{n-1} E$ is injective, so the first-polar map is defined on all of $\mathbb{P}(E^\vee)$. This is exactly the complement of the rank-one boundary: after a change of coordinates, a nonzero kernel vector is ∂_X, and $\partial_X f = 0$ forces f

to be a scalar multiple of $Y^{[n]}$. Such a point lies on the normal rational curve and is already excluded from the split-free locus. Then

$$\Phi_f(\mathbb{P}(E^\vee)) \subset \{b : \operatorname{rank} C_b^{(2)} \leq 2\} \quad \Longleftrightarrow \quad \operatorname{rank} A \leq 2$$

scheme-theoretically and in every characteristic. If $\operatorname{rank} A = 3$, the rank-drop parameters form a subscheme of $\mathbb{P}(E^\vee)$ of degree at most three.

Proof. For a nonzero $\lambda \in E^\vee$, multiplication identifies $\operatorname{Sym}^{n-3} E^\vee$ with the hyperplane

$$H_\lambda = \lambda \operatorname{Sym}^{n-3} E^\vee \subset \operatorname{Sym}^{n-2} E^\vee,$$

and functoriality gives

$$C^{(2)}_{\iota_\lambda f} = A|_{H_\lambda}.$$

The forward implication is immediate. Conversely, suppose $\operatorname{rank} A = 3$ and put $K = \ker A$, so $\dim K = n-4$. If the lower rank is at most two, rank–nullity on the $(n-2)$-dimensional space H_λ gives

$$\dim(K \cap H_\lambda) \geq (n-2) - 2 = n - 4 = \dim K.$$

Thus $K \subset H_\lambda$. Containment for every geometric $[\lambda]$ would imply

$$K \subset \bigcap_{[\lambda] \in \mathbb{P}(E^\vee_{\bar{k}})} \lambda \operatorname{Sym}^{n-3} E^\vee_{\bar{k}} = 0,$$

because a nonzero binary form cannot be divisible by every linear form. This contradicts $\dim K = n - 4 > 0$.

The maximal minors of $C^{(2)}_{\iota_\lambda f}$ are homogeneous cubics in λ. Vanishing at every geometric point is therefore identical vanishing, proving the scheme-theoretic assertion. In the non-contained case at least one cubic is nonzero, and the common rank-drop scheme lies in its degree-three zero divisor. $\square$

Thus Proposition 5.21 covers the entire non-rank-boundary locus on which the first-polar line is defined; the excluded noninjective locus is the rank-one normal-rational-curve locus just identified. After rank one and rational split secants are removed as shallow, the upper rank-two locus consists of the persistent tangent and conjugate-secant families. On a persistent fiber the nonsplit torus acts by $z \mapsto z^n$, giving T/T^n modulo inversion and Frobenius, while the tangent unipotent cocycle is $z \mapsto z + nu$.

Modular loci are equally explicit. If $\mathcal{N}$ is a nucleus in the lower divided-power module, Definition 5.1 identifies the complete space whose first-polar line lies in $\mathcal{N}$ with the linear kernel scheme $\mathcal{M}_n(\mathcal{N})$. Lucas' theorem computes the nucleus coordinates, and this kernel computes every coherent lift without an ambient syndrome census. At redundancy r, if $p > r-1$, Lucas' theorem leaves no nucleus coordinate, so $\mathcal{M}_r$ is empty. This removes the modular term only; the R5 cyclic carrier survives in large characteristic.

Proposition 5.22 (Higher Lucas endpoint test)**.** Let $q = 2^m$, let $d = 2^s$ with $s \geq 2$, and consider the degree-$(d+1)$ coherent Lucas carrier

$$\mathbb{P}\langle e_2, \ldots, e_{d-1}\rangle \subset \mathbb{P}(\Gamma^{d+1} E).$$

At its distinguished endpoint e_{d-1}, the Hankel kernel $W_{e_{d-1}} \subset \operatorname{Sym}^d E^\vee$ contains the canonical section

$$\mathcal{U}_s = \langle 1, t, t^d \rangle.$$

This section contains a completely split squarefree polynomial of degree d over $\mathbb{F}_q$ if and only if $s \mid m$.

The case $s = 2$ is the binary nucleus already occurring at the first two polar levels. For the first higher case $s = 3$, the conclusion for the full kernel is strictly stronger: for every $m \geq 3$, the entire $\mathrm{PGL}_2(q)$-orbit of $e_7 \in \mathbb{P}(\Gamma^9 E)$ is shallow.

Proof. The subfield-divisibility criterion in the first paragraph is equivalently the characteristic-two, $k = 2^s+1$ projective-subline case of [WWH26, Prop. 11]. We give the endpoint-coordinate proof because the identification with the present Hankel kernel, and the stronger full e_7-orbit conclusion, are needed here. Lucas' binomial criterion [GH00, Lem. 1 and Thm. 1], applied to the two consecutive contraction rows, gives the displayed carrier and leaves $1, t, t^d$ at e_{d-1}. More explicitly, its two Hankel equations are

$$b_{d-2} = b_{d-1} = 0,$$

so

$$W_{e_{d-1}} = \langle 1, t, \dots, t^{d-3}, t^d \rangle \subset \mathrm{Sym}^d E^\vee.$$

A squarefree degree-d member of $\mathcal{U}_s$ has the form

$$Ct^d + Bt + A, \qquad BC \neq 0.$$

Over an algebraic closure, its roots form an affine translate of $\beta\mathbb{F}_d$, where $\beta^{d-1} = B/C$. If all roots lie in $\mathbb{F}_q$, ratios of two nonzero root differences recover $\mathbb{F}_d$; hence $\mathbb{F}_d \subseteq \mathbb{F}_q$, equivalently $s \mid m$. Conversely, when $s \mid m$, the polynomial $t^d + t$ is split and squarefree over $\mathbb{F}_q$.

For $s = 3$, the two endpoint Hankel equations give

$$W_{e_7} = \langle 1, t, t^2, t^3, t^4, t^5, t^8 \rangle.$$

Choose any three-dimensional $\mathbb{F}_2$-subspace $V \subseteq \mathbb{F}_q$. Its subspace polynomial is

$$L_V(t) = \prod_{v \in V} (t - v) = t^8 + A_4 t^4 + A_2 t^2 + A_1 t, \qquad A_1 = \prod_{v \in V \setminus \{0\}} v \neq 0.$$

Thus $L_V \in W_{e_7}$ is completely split and squarefree. For the contragredient apolar action,

$$gW_f = W_{gf}.$$

The same action preserves split squarefree root divisors, so the witness transports throughout the orbit. □

Remark 5.23 (Scope of the contained calculations). The modular contraction kernel is an exact finite linear-algebra construction. Proposition 5.21 supplies the ordinary persistent component uniformly, and Lemma 5.7 supplies the collision alternative. Theorem 1.3 settles the remaining arbitrary-level question: no other recursively pointed contained component occurs. Proposition 5.22 settles the canonical section and first fresh endpoint orbit, not the other arithmetic strata of a higher Lucas carrier.

6 Redundancies six and seven: complete and gated results

6.1 Redundancy six

For

$$f = (a_0 : \cdots : a_5) \in \mathbb{P}(\Gamma^5 E), \qquad W_f = \ker \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \end{pmatrix},$$

the system W_f is a net of quartics off the quintic normal rational curve. It is split-free exactly when it has no completely split squarefree quartic.

Proposition 6.1 (Persistent net and orbit law)**.** If $\gcd(W_f) = Q$ has degree two, then

$$W_f = Q \operatorname{Sym}^2 E^\vee.$$

The split-free persistent locus has size $q(q+1)^2/2$. On the tangent fiber the unipotent coordinate is $x \mapsto x + 5u$; on the irreducible quadratic fiber the norm-one torus acts through $z \mapsto z^5$, with inversion and coefficient Frobenius acting on T/T^5.

Proof. For fixed Q, the syndromes annihilating $Q \operatorname{Sym}^3 E^\vee$ form a projective line. If Q is irreducible, all $q+1$ points remain; if $Q = \lambda^2$, one endpoint has dependent Hankel rows, leaving q points. Hence

$$q(q+1) + \frac{q(q-1)}{2}(q+1) = \frac{q(q+1)^2}{2}.$$

Normalize a square to U^2. The nondegenerate fiber is $[e_4 + xe_5]$, and direct degree-five substitution gives $x \mapsto x + 5u$. For an irreducible Q, its stabilizer identifies the $q+1$-point fiber with $T = \{z \in \mathbb{F}_{q^2}^\times : z^{q+1} = 1\}$ and acts by fifth powers; the normalizer inverts z, while coefficient Frobenius multiplies T/T^5 by the characteristic. This proves the stated orbit law and the characteristic-five tangent splitting. □

Lemma 6.2 (Exact linear gcd is shallow)**.** Let $q \geq 7$. If $\gcd(W_f)$ has degree one, then W_f contains a split squarefree quartic.

Proof. Write $W_f = LV$, where L is rational and $V \subset \operatorname{Sym}^3 E^\vee$ has vector dimension three. It suffices to find in V a product of three distinct rational factors, none equal to L. Move the root of L to infinity and let ℓ span $V^\perp$. On the cubic with distinct finite roots x, y, z, write

$$\ell\big((T - xU)(T - yU)(T - zU)\big) = c_0 + c_1(x + y + z) + c_2(xy + xz + yz) + c_3xyz =: P(x, y, z).$$

Suppose that this value is nonzero for every distinct triple. For fixed distinct x, y, write $P = A(x, y) + zB(x, y)$, where

$$B(x, y) = c_1 + c_2(x + y) + c_3xy.$$

If $B(x, y) \neq 0$, the unique zero in z must be x or y. Consequently

$$B(x, y)P(x, y, x)P(x, y, y) = 0$$

for every $x \neq y$; it also vanishes when $B(x, y) = 0$. This polynomial has degree at most four in each variable. Since $q - 1 > 4$, first varying $y \neq x$ and then x shows that it is the zero polynomial. The polynomial ring is a domain. Neither $P(x, y, x)$ nor $P(x, y, y)$ vanishes identically unless $\ell = 0$, as comparison of the coefficients of $1, x, y, x^2, xy, x^2y$ shows. Hence $B = 0$, so $c_1 = c_2 = c_3 = 0$.

Thus V is the hyperplane of cubics vanishing at infinity, and every member of V is divisible by L. This would make $\gcd(W_f)$ divisible by L^2, contrary to its exact degree. Therefore a required cubic exists, and multiplying it by L gives a split squarefree member of W_f. □

Proposition 6.3 (First-polar line and secant degree)**.** For $r \in \mathbb{F}_q$ put

$$b(r) = (a_1 - ra_0, \ldots, a_5 - ra_4), \qquad b(\infty) = (a_0, \ldots, a_4).$$

Then

$$(T - rU)g \in W_f \iff g \in W_{b(r)}.$$

If W_f has trivial gcd, its polar line meets the lower secant cubic in degree at most three.

Proof. The first equivalence follows by expanding the two Hankel equations of $(T - rU)g$. Let

$$C_f = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \\ a_2 & a_3 & a_4 & a_5 \end{pmatrix}.$$

The lower catalecticant equals

$$C_f \begin{pmatrix} -r & 0 & 0 \\ 1 & -r & 0 \\ 0 & 1 & -r \\ 0 & 0 & 1 \end{pmatrix}.$$

Cauchy–Binet gives the secant determinant

$$D_f(r) = -\Delta_{012}r^3 + \Delta_{013}r^2 - \Delta_{023}r + \Delta_{123}.$$

It vanishes identically exactly when $\operatorname{rank} C_f \leq 2$, equivalently when W_f has quadratic gcd. In the trivial-gcd case it therefore has at most three zeros. ☐

Proposition 6.4 (Cyclic and collision degrees)**.** For a trivial-gcd net:

(i) in characteristic different from two and three, the lower tame cyclic carrier is a binomially rescaled Veronese surface of degree four and contains no polar line;

(ii) the pointed ramification condition has degree at most six and does not vanish identically;

(iii) an S_3 lower pencil with the marked root deleted has total deletion degree at most nineteen.

Proof. In syndrome coordinates the tame cyclic surface is

$$\left\{\operatorname{diag}(1,4,6,4,1)^{-1}[Q^2] : [Q] \in \mathbb{P}(\operatorname{Sym}^2 E^\vee)\right\}.$$

When 2 and 3 are invertible this is the Veronese surface, of degree four, and contains no line. Characteristic three has a different wild closure, treated in Proposition 6.5. For (ii), Lemma 5.7 with $j = 6$ gives the degree-six bound except possibly in characteristic two. In that characteristic, inseparability would identify W_f, after a change of coordinates, with the full pure-square space $\langle T^4, T^2U^2, U^4\rangle$. Its annihilator is $\langle e_1^\vee, e_3^\vee\rangle$. For the two consecutive Hankel rows this would require simultaneously

$$a_0 = a_2 = a_4 = 0, \qquad a_1 = a_3 = a_5 = 0,$$

contrary to $f \neq 0$. Thus the bound holds in every characteristic.

Finally the R5 off-diagonal curve has deletion degree thirteen, including its possible rational singular point. The unique cubic through a specified marker contributes at most six ordered pairs, giving $(g, \delta, \kappa) = (1, 19, 1)$ and

$$q + 1 - 2\sqrt{q} > 19.$$

Its first prime-power solution is 29. ☐

Proposition 6.5 (Modular contained components)**.** In characteristic three the lower wild carrier is the degree-four cone

$$\operatorname{Join}(e_2, C_4),$$

and no polar line of a trivial-gcd net is contained in it. In characteristic two the cyclic carrier is the plane

$$\Pi_{\mathrm{cyc}} = \mathbb{P}\langle e_1, e_2, e_3\rangle,$$

whose unique contained polar-line component is

$$\mathcal{N}_3 = \mathbb{P}\langle e_2, e_3\rangle.$$

Proof. Every line on the wild cone is a ruling. By equivariance it suffices to compare the two consecutive rows on one ruling; in the frame $\langle e_2, e_4\rangle$ the conditions

$$(a_0, \ldots, a_4),\ (a_1, \ldots, a_5) \in \langle e_2, e_4\rangle$$

force the overlapping coordinates successively to vanish. Undoing the frame gives $f = \mu\nu(t)$, including $f = \mu e_5$ at infinity. These are fixed-factor/rank-boundary syndromes, so no trivial-gcd polar line is contained in the wild cone.

In characteristic two, direct substitution gives

$$g \cdot e_2 = [0 : \delta\gamma : \alpha\delta + \beta\gamma : \alpha\beta : 0],$$

whose closure is Π_{cyc}. The consecutive-row overlap is now literal: the first row lies in the plane exactly when $a_0 = a_4 = 0$, and the second exactly when $a_1 = a_5 = 0$. Hence a polar line lies in the plane exactly when

$$a_0 = a_1 = a_4 = a_5 = 0,$$

namely when $f \in \mathbb{P}\langle e_2, e_3\rangle$. This proves both existence and uniqueness of the contained binary component. $\square$

Proposition 6.6 (Binary nucleus arithmetic)**.** The scheme $\mathcal{N}_3(\mathbb{F}_q)$ is one projective-semilinear orbit of $q + 1$ points. It is split-free over $q = 2^m$ exactly when m is odd.

Proof. For the representative e_3,

$$W_{e_3} = \langle 1, t, t^4\rangle.$$

If $C = 0$, the member $A+Bt+Ct^4$ has degree at most one and cannot have four distinct roots. Assume $C \neq 0$. The root differences of $A+Bt+Ct^4$ then lie in the kernel of $u \mapsto u^4+(B/C)u$. Four distinct rational roots exist exactly when all roots of $u^3 = B/C$ are rational, equivalently when $3 \mid q - 1$, which for $q = 2^m$ means m is even. The stabilizer of e_3 is the Borel of order $q(q-1)$, so the orbit has $q+1$ points; its lower-unipotent transforms and endpoint are precisely $\mathcal{N}_3(\mathbb{F}_q)$. $\square$

Proposition 6.7 (Exhaustive R6 contained-component classification)**.** For a redundancy-six syndrome, the alternatives in Definition 5.13 are exhausted as follows:

(i) a noninjective contraction map is the rank-one normal-rational- curve boundary and is shallow;

(ii) a polar line contained in the lower secant cubic comes from the upper rank-two catalecticant locus; after its shallow rational secants are removed, this is the persistent tangent and conjugate-secant locus;

(iii) no trivial-gcd polar line is contained in the tame cyclic surface or the characteristic-three wild cone;

(iv) in characteristic two, containment in the cyclic plane occurs exactly for $f \in \mathcal{N}_3 = \mathbb{P}\langle e_2, e_3\rangle$;

(v) on the remaining trivial-gcd locus the collision divisor is not identically zero.

The exact-linear-gcd lower stratum is not a contained component: it is the separate quadratic-graph package of Proposition 4.3. Consequently the complete nonshallow contained list is the persistent locus together with $\mathcal{N}_3$ in characteristic two. There is no unlisted contained component.

Proof. The noninjective and secant-cubic alternatives are Proposition 5.21 and Proposition 6.3. The tame cyclic and collision assertions are Proposition 6.4; the wild and binary specializations are Proposition 6.5. These are all the clauses of the terminal carrier in Definition 5.13: contraction indeterminacy, containment in each component of the lower bad scheme, and identical collision. Hence the list is exhaustive. □

Proposition 6.8 (R6 one-step exhaustion)**.** Assume $q \geq 29$. Then every split-free redundancy-six syndrome lies in the persistent quadratic-gcd locus or, when $q = 2^m$ with odd m, in $\mathcal{N}_3(\mathbb{F}_q)$.

Proof. The complete contained branch is Proposition 6.7. The common-factor trichotomy is exhaustive. A quadratic gcd gives the persistent tangent and conjugate-secant loci, together with the shallow rational secant; an exact linear gcd is shallow by Lemma 6.2. It remains to treat a trivial-gcd net.

Its first-polar line is not contained in the lower secant cubic, by Proposition 6.3. The pullback of that cubic has degree at most three. The tame cyclic surface has degree four and contains no polar line. In characteristic three its wild replacement has the same degree and no contained trivial-gcd polar line; in characteristic two the replacement is a plane, whose only contained component is $\mathcal{N}_3$, by Proposition 6.5. Finally, Proposition 6.4 gives a nonzero collision divisor of degree at most six. Outside the contained binary component, the excluded parameter scheme therefore has degree

$$d \leq 3 + 4 + 6 = 13.$$

For every remaining marker, the lower cubic pencil has exact linear gcd or geometric monodromy S_3. In the first case the graph argument of Proposition 4.3 is pointed as follows: besides its eight original deletions, remove the at most two graph points whose split quadratic contains the new marker. Since $q + 1 > 10$, a split squarefree cubic avoiding that marker remains. Here the fixed gcd λ_0 differs from the marker r. Indeed, $\lambda_0 = r$ says that every cubic in $W_{b(r)}$ vanishes at r, so every lifted quartic has a double root there; by Lemma 5.6, this is exactly a point of the degree-six collision divisor already placed in Z_f, so no additional deletion budget is needed. In the S_3 case, Lemma 4.6 supplies the geometrically integral off-diagonal curve. Its singular, diagonal, and branch deletions have degree at most thirteen, and deleting the unique cubic through the marker removes at most six ordered pairs. Thus

$$(g, \delta, \kappa) = (1, 19, 1), \qquad q + 1 - 2\sqrt{q} > 19$$

for every $q \geq 29$. Explicitly,

$$\mathcal{H}_1(1, 19) = 1 + \left\lfloor (1 + \sqrt{19})^2 \right\rfloor = 29, \qquad 30 - 2\sqrt{29} > 19.$$

These data form the one-step package of Definition 5.4, with $d = 13$, so Theorem 5.17 produces a split squarefree quartic.

This is one field-order threshold, not three characteristic-dependent estimates: the first characteristic-three and characteristic-two prime powers above it happen to be 81 and 32, respectively. On $\mathcal{N}_3$, Proposition 6.6 gives the asserted odd-degree split-free orbit and proves shallowness in even extension degree. □

The covering-radius gate is complete. Its exact high-rate endpoint is $q = 8$, since $6 = \lfloor 8/2 \rfloor + 2$. Thus Seroussi–Roth nonextendability and Dür's equivalence give $\rho(\mathrm{PRS}(q-5)) = 5$ for $q \geq 8$, while Certificate R6 checks it directly at $q = 7$ and independently confirms $q = 8$.

Theorem 6.9 (Redundancy six)**.** For every prime power $q \geq 7$, the deep syndromes of $\mathrm{PRS}(q-5)$ are the persistent locus of size $q(q+1)^2/2$, together with:

(i) respectively $18, 11, 4, 2, 1$ exceptional $\mathrm{PGL}_2(q)$ orbits at $q = 7, 8, 9, 11, 13$;

(ii) one characteristic-two nucleus orbit of size $q + 1$ when $q = 2^m$ and odd $m \geq 5$.

The lower bound $m \geq 5$ avoids double counting: the remaining odd case $m = 3$, namely $q = 8$, is already the $|\mathcal{O}| = 9$ row of the exceptional table. There are no other exceptional orbits. The complete persistent orbit law is T/T^5 modulo inversion and Frobenius, with a tangent fixed/nonzero split exactly in characteristic five. At $q = 7, 8, 9, 11, 13$, the exceptional direction totals are

$$5152, \quad 4713, \quad 1800, \quad 792, \quad 546,$$

and the corresponding total numbers of deep syndrome directions are

$$5376, \quad 5037, \quad 2250, \quad 1584, \quad 1820.$$

Proof. Propositions 6.7 and 6.8 give the complete conceptual exhaustion for every prime power $q \geq 29$. Certificate R6 exhausts

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27\},$$

finds exceptional projective-orbit counts $18, 11, 4, 2, 1$ only at the five displayed fields, and no further orbit at the remaining fields. Together with the separate radius gate, this covers every prime power $q \geq 7$. □

The listed sources are disjoint. The persistent locus has quadratic gcd, every finite exceptional row has trivial gcd, and the separate nucleus term occurs only for binary fields beyond the finite table.

The five small tables are exhaustive computations. For an exceptional net W_f, let $I_f(r)$ be the number of irreducible cubic members of the quotient pencil $W_{b(r)}$, including the infinity chart, and define its shared-root pair count by

$$E_f = \sum_{r \in \mathbb{P}^1(\mathbb{F}_q)} \binom{I_f(r)}{2}.$$

In odd characteristic let

$$F_f(T, U) = \sum_{i=0}^{5} a_i T^{5-i} U^i$$

be the binary quintic attached to $f = (a_0 : \cdots : a_5)$, and let $\tau_5(f)$ be its rational factorization type; for example, $1^3 + 2$ denotes a triple rational factor and an irreducible quadratic factor. Their semilinear orbit counts are $18, 5, 2, 2, 1$; E_f, together in odd characteristic with $\tau_5(f)$, separates the classes up to precisely Frobenius fusion. Table 7 prints the canonical representatives, stabilizers, orbit sizes, and fusion lengths.

Table 7: Exceptional redundancy-six classes. Each row is one projective-semilinear class. The columns give a canonical syndrome representative, the size and stabilizer order of each constituent $\mathrm{PGL}_2(q)$-orbit, the number ϕ of such orbits fused by coefficient Frobenius, the shared-root pair count E, and the binary-quintic factor type τ_5 in odd characteristic.

q	representative	$\lvert\mathcal{O}\rvert$	$\lvert G_f\rvert$	ϕ	E	τ_5
7	$[0:0:0:1:0:1]$	168	2	1	7	1^3+2
7	$[0:0:1:0:3:1]$	336	1	1	19	1^2+1+2
7	$[0:0:1:0:3:2]$	336	1	1	15	1^2+3
7	$[0:1:0:0:0:1]$	168	2	1	10	$1+2+2$
7	$[0:1:0:0:2:0]$	112	3	1	33	$1+1+3$
7	$[0:1:0:1:1:2]$	336	1	1	9	$1+4$
7	$[0:1:0:1:1:5]$	336	1	1	9	$1+1+3$
7	$[0:1:0:1:3:0]$	336	1	1	21	$1+1+1+2$
7	$[0:1:0:1:3:3]$	336	1	1	7	$1+1+3$
7	$[0:1:0:1:3:6]$	336	1	1	14	$1+1+3$
7	$[0:1:0:3:0:5]$	168	2	1	11	$1+4$
7	$[0:1:0:3:0:6]$	168	2	1	20	$1+4$
7	$[0:1:0:3:2:1]$	336	1	1	13	$1+1+3$
7	$[1:0:0:0:1:2]$	336	1	1	7	$1+4$
7	$[1:0:0:0:1:3]$	336	1	1	23	5
7	$[1:0:0:3:1:3]$	336	1	1	13	$1+4$
7	$[1:0:1:0:5:2]$	336	1	1	8	$1+4$
7	$[1:0:1:0:6:2]$	336	1	1	8	5
8	$[0:0:0:1:0:0]$	9	56	1	0	–
8	$[0:1:1:0:2:1]$	504	1	3	20	–
8	$[0:1:1:0:2:5]$	504	1	3	19	–
8	$[1:0:0:1:3:6]$	504	1	3	14	–
8	$[1:0:1:3:5:2]$	168	3	1	54	–
9	$[0:1:0:0:0:4]$	180	4	2	28	$1+4$
9	$[0:1:0:1:4:2]$	720	1	2	29	$1+4$
11	$[0:0:0:1:0:0]$	132	10	1	60	1^3+1^2
11	$[0:1:0:2:0:10]$	660	2	1	30	$1+4$
13	$[0:1:0:0:0:2]$	546	4	1	60	$1+4$

For $q=8$, integers are binary polynomial-basis codes with $\alpha^3=\alpha+1$; for $q=9$, they are ternary polynomial-basis codes with $\alpha^2=-1$. Thus $\sum\phi=(18,11,4,2,1)$ and $\sum\phi|\mathcal{O}|=(5152,4713,1800,792,546)$, in the field order $7,8,9,11,13$.

6.2 Redundancy seven

For

$$f=(a_0:\ldots:a_6)\in\mathbb{P}(\Gamma^6E),\qquad W_f=\ker\begin{pmatrix}a_0&a_1&a_2&a_3&a_4&a_5\\a_1&a_2&a_3&a_4&a_5&a_6\end{pmatrix},$$

the system W_f is a four-dimensional web of quintics. For an affine marker r the contractions are

$$b(r)=(a_1-ra_0,\ldots,a_6-ra_5).$$

Coefficient collection gives

$$(t - r)g \in W_f \quad \iff \quad g \in W_{b(r)};$$

the lift is squarefree exactly when $g(r) \neq 0$.

The contracted net $W_{b(r)}$ has quadratic gcd exactly when its three-row catalecticant has rank at most two. Unless the first-polar line is contained in that rank-two locus, these parameters form the degree-at-most-three intersection of Proposition 6.3; the contained case is Corollary 6.14.

Proposition 6.10 (Complete two-marker lower package)**.** Let W_h be a trivial-gcd quartic net and let x be a prescribed forbidden root. Assume in characteristic two that $h \notin \mathcal{N}_3$. For every prime power $q \geq 37$, the net contains a split squarefree quartic avoiding x.

Proof. By equivariance choose coordinates in which x is finite; all parameter equations below are then homogenized in s, so the infinity chart is included. Choose the second contraction marker s. The unavailable values of s comprise: $s = x$, of degree one; at most three intersections with the lower secant cubic, by Proposition 6.3; at most four intersections with the cyclic surface or its characteristic-three wild replacement; and at most six points of the collision divisor, by Propositions 6.4 and 6.5. In characteristic two the cyclic replacement is a plane, so its pullback has degree at most one unless the polar line is contained in it, the excluded case $h \in \mathcal{N}_3$, by Proposition 6.5. We must also exclude the parameters for which the lower pencil has fixed factor x. They are cut out by the 3-minors obtained by appending the fixed evaluation row $(1, x, x^2, x^3)$ to the two Hankel rows of $b(s)$, and hence have degree at most two in s. These minors are not all zero. If they vanished identically, then

$$W_h \cap \{g : g(s) = 0\} \subseteq \{g : g(x) = 0\}$$

for every geometric s. Given $g \in W_h$, choose a geometric root s of g; then $g(x) = 0$. Thus x would divide every member of W_h, contradicting the trivial-gcd hypothesis. Hence the complete second-step parameter scheme has degree at most

$$1 + 3 + 4 + 6 + 2 = 16 < q + 1.$$

Choose s outside it.

The resulting cubic pencil is neither persistent nor cyclic. If it has exact linear gcd λ_0, then $\lambda_0 \notin \{x, s\}$: equality with x is excluded by the fixed-factor minors above, while equality with s is the pointed collision excluded by Proposition 6.4. The graph proof of Proposition 4.3 has eight original deletions; avoiding the two additional markers costs at most two graph points each. Since $q + 1 > 12$, a split squarefree cubic avoiding x and s remains.

Otherwise the bottom cover has geometric monodromy S_3. Its off-diagonal curve is geometrically integral of arithmetic genus one. The singular, diagonal, and branch deletions cost at most thirteen, and the fibers through x and s cost at most six ordered pairs each. Hence

$$(g, \delta, \kappa) = (1, 25, 1), \qquad q + 1 - 2\sqrt{q} > 25,$$

whose first prime-power solution is 37. In either case the cubic lifts through s to a split squarefree quartic in W_h avoiding x. $\square$

Proposition 6.11 (Exact-gcd-one avoidance)**.** Suppose the lower net is $\ell_x U$, where $U = \ker \lambda \subset \operatorname{Sym}^3 E^\vee$ has dimension three. If $x = r$, the parameter belongs to the collision divisor. If $x \neq r$ and $q \geq 16$, then U contains a split cubic avoiding both x and r.

Proof. If $x = r$, every lifted quintic has the repeated factor $(t-r)^2$. Lemma 5.6 identifies this with the degree-eight collision divisor of Proposition 6.12. This is the unavailable branch, so assume $x \neq r$. Put $S = \mathbb{P}^1(\mathbb{F}_q) \setminus \{x, r\}$. For ordered distinct $u, v \in S$, the map $w \mapsto \lambda(\ell_u \ell_v \ell_w)$ is linear. If it vanishes identically, choose any remaining w; otherwise it has one rational zero. The equations $w = x, r, u, v$ have respective bidegrees $(1,1), (1,1), (2,1), (1,2)$ in (u, v). None is identically zero: the first two would add a common factor, and the last two are excluded because double-root cubics span $\operatorname{Sym}^3 E^\vee$. A nonzero bidegree-$(a, b)$ divisor on $\mathbb{P}^1 \times \mathbb{P}^1$ has at most $(a+b)(q+1)$ rational points. Thus the bad pairs number at most $10(q+1)$, whereas $(q-1)(q-2) > 10(q+1)$ for $q \geq 16$. □

Proposition 6.12 (Collision divisor)**.** After fixed factors are removed, the collision divisor of the moving g^3_5 has degree at most eight.

Proof. This is Lemma 5.7 with $j = 7$: the series is separable and $2j - 6 = 8$. □

The characteristic-two nucleus line at redundancy six has exactly one coherent lift, the central point e_3, with

$$W_{e_3} = \langle 1, t, t^4, t^5 \rangle.$$

It is deep for odd m; for even m, the homogenization of $t^4 + t$ has the five roots $\mathbb{P}^1(\mathbb{F}_4)$.

Proposition 6.13 (Central binary lift)**.** The complete inverse image of the R6 nucleus line under consecutive sextic contraction is $[e_3]$. It is split-free over $\mathbb{F}_{2^m}$ exactly when m is odd.

Proof. Putting both consecutive rows in $\mathbb{P}\langle e_2, e_3 \rangle$ gives $a_0 = a_1 = a_2 = a_4 = a_5 = a_6 = 0$. Thus only e_3 remains, with $W_{e_3} = \langle 1, t, t^4, t^5 \rangle$. For odd m, contraction at a root of a hypothetical split quintic would give a split quartic on the R6 nucleus line, contradicting Proposition 6.6. For even m, $\mathbb{F}_4 \subset \mathbb{F}_q$ and the homogenization of $t^4 + t$ has its four affine $\mathbb{F}_4$ roots and the simple root at infinity. □

Corollary 6.14 (Exhaustive R7 contained-component classification)**.** The assertion CC$(6,1)$ is exhaustive in every characteristic. A noninjective contraction is the shallow rank-one boundary. Containment in the lower persistent locus gives upper catalecticant rank at most two; after the shallow rational secants are removed, these are exactly the persistent tangent and conjugate-secant families. The complete lift of the only additional lower component, the characteristic-two nucleus line, is the central point e_3. The remaining collision divisor is nonzero. Thus the nonshallow contained list is precisely the persistent locus and, in characteristic two, e_3; there is no further contained component.

Proof. Apply Proposition 5.21 with $n = 6$. The three rational types of the quadratic recurrence are split squarefree, repeated rational, and irreducible. They give respectively the shallow rational secant, tangent, and conjugate-secant loci. The remaining lower binary nucleus line has the separate complete lift computed in Proposition 6.13. Proposition 6.12 shows that the collision divisor is nonzero. These are respectively the contraction-indeterminacy, lower-component-containment, and identical-collision alternatives of Definition 5.13, so the list is exhaustive. □

Theorem 6.15 (Redundancy-seven classification)**.** Certificate R7 unconditionally classifies every prime power $q \leq 32$ in its displayed domain. For every prime power $q \geq 13$, the split-free syndromes at redundancy seven are the persistent locus of size $q(q+1)^2/2$, together with the central nucleus point when $q = 2^m$ and m is odd. At $q = 7, 8, 9, 11$ the additional split-free orbit-size profiles, after removing the persistent locus and also the central nucleus singleton at $q = 8$, are:

$$\begin{array}{r|l} 7 & 56,\ 84^5,\ 112^2,\ 168^{45},\ 336^{141} \\ 8 & 63,\ 72,\ 84^3,\ 168^4,\ 252^{24},\ 504^{86} \\ 9 & 180^3,\ 240^6,\ 360^{18},\ 720^{27} \\ 11 & 264^2,\ 440,\ 660^2. \end{array}$$

This is the complete all-field split-free classification. Since Seroussi–Roth nonextendability and Dür's equivalence give $\rho(\mathrm{PRS}(q-6)) = 6$ for every prime power $q \geq 11$, it is also the complete deep-hole classification in that range. At $q = 7, 8, 9$ the table is not promoted to a code deep-hole classification without a separate covering-radius calculation. The persistent orbit law is T/T^6 modulo inversion and Frobenius, with tangent splitting in characteristics two and three.

Proof. For fixed irreducible quadratic Q, the persistent fiber has $q+1$ points, while for $Q = \lambda^2$ it has q non-rank-one points. There are $q(q-1)/2$ irreducible quadratics and $q+1$ rational squares. Hence the persistent count is

$$\frac{q(q-1)}{2}(q+1) + (q+1)q = \frac{q(q+1)^2}{2}.$$

For $q \geq 37$, choose a first marker outside the at most three catalecticant intersections, eight collision parameters, and, in characteristic two, one intersection with the nucleus line. If the contracted quartic net has trivial gcd, Proposition 6.10 supplies the complete second one-step package: it excludes the cyclic/wild carrier and fixed-gcd/marker coincidences before applying either the pointed linear-gcd graph or the S_3 cover. If the contracted net has exact linear gcd, Proposition 6.11 applies; its excluded $x = r$ case is already among the eight collision parameters. Corollary 6.14 leaves exactly the persistent locus and Proposition 6.13. Certificate R7 covers

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32\}$$

by an independent five-secant replay. The radius theorem promotes the split-free list only from $q \geq 11$. □

For the four finite rows Certificate R7 contains the canonical representative, stabilizer, persistent/central flags, and Frobenius link of every orbit. The displayed size profile is only a compact summary; repeated sizes are distinguished by those canonical records. There are respectively $194, 119, 54, 5$ exceptional representatives at $q = 7, 8, 9, 11$; the complete lists are printed in `supplement/CLASSIFICATION-RECORDS.md`, while the matching JSON is the machine-readable record. The following gives one printed canonical representative for every orbit-size block in the displayed profile; repeated orbits within a block are distinguished in the full list:

q	$\lvert\mathcal{O}\rvert$	representative
7	56	$[0:0:0:0:1:0:0]$
	84	$[0:1:0:0:0:1:0]$
	112	$[0:0:1:0:0:2:0]$
	168	$[0:0:0:0:1:0:1]$
	336	$[0:0:0:1:0:1:1]$
8	63	$[0:0:0:1:0:0:1]$
	72	$[0:0:0:0:1:0:0]$
	84	$[0:1:1:1:1:3:6]$
	168	$[1:0:1:1:3:7:7]$
	252	$[0:0:1:0:1:0:0]$
	504	$[0:0:1:0:1:0:1]$
9	180	$[0:1:0:0:0:4:0]$
	240	$[0:0:1:0:1:1:1]$
	360	$[0:1:0:1:3:3:2]$
	720	$[0:0:1:1:0:2:2]$
11	264	$[0:1:0:0:0:0:2]$
	440	$[1:0:1:1:4:3:3]$
	660	$[1:0:1:0:1:4:8]$.

For $q = 8, 9$, the coordinates use the same polynomial-basis encoding specified below Table 7.

The $q = 19$ lower net

$$W = \langle 1, t^3, t^4 \rangle$$

shows why the parameter must remain marked. Its split squarefree members are exactly

$$U(T^3 - uU^3), \qquad u \in (\mathbb{F}_{19}^{\times})^3 = \{1, 7, 8, 11, 12, 18\}.$$

All six contain the marked point at infinity. The obstruction is pointed rather than intrinsic to W: at the marker 0, for example, $U(T^3 - U^3)$ is an admissible member. No coherent sextic polar line lies in the infinity-pointed orbit. Thus the R7 proof escapes over the marker parameter; classifying unpointed bad lower fibers would retain a false exception.

Table 8: Field-range closure ledger. "Certificate" means a complete finite classification with canonical representatives, not merely an orbit count.

Result	prime powers	closure mechanism
R5	$7, 8, 9, 11, 13, 16, 17, 19$	direct census and independent replay
	$q \geq 23$	cubic-cover geometry, Lemma 4.6
R6	$7, 8, 9, 11, 13, 16$	direct syndrome/radius census
	$17, 19, 23, 25, 27$	finite structural bridge certificate
	$q \geq 29$	marked-cover point bound and contained components
R7	$7, 8, 9, 11$	direct split-free census
	$13, 16, 17, 19, 23, 25, 27, 29, 31, 32$	coherent-polar finite bridge certificate
	$q \geq 37$	marked-cover point bound and contained components

The R7 rows classify split-free syndromes. Conversion to code deep-hole directions uses the separate covering-radius gate and is made only for $q \geq 11$. The public certificate schema and exact replay boundaries are given in the supplement; the table does not promote orbit-size profiles to identifiers.

There is no prime power in any of the open numerical intervals $(19, 23)$, $(27, 29)$, or $(32, 37)$. Thus adjacent finite and geometric rows in Table 8 leave no unlisted field order.

7 Verification and formal boundary

No geometric classification theorem in this paper is claimed to be formalized in Lean. The formal layer checks coordinate identities and conditional synthesis implications with the component geometry, rational-point existence, group actions, and certificate semantics kept as explicit inputs.

The conceptual theorems above are accompanied by deterministic evidence bundles. Table 9 records which claims use exhaustive computation. Each public label resolves, through the supplement schema, to its generator, certificate, replay where stated, and SHA-256 manifest. Table 1 gives the corresponding claim-level mathematical dependencies.

Table 9: Public verification map. Exact filenames, commands, hashes, domains, and stop conditions belong to the electronic supplement.

Public label	Verified domain	Replay and trust boundary
Certificate R5	nineteen audited field records; eight-field finite bridge below 23	independent pointwise and orbitwise replay; canonical representatives and exhaustion records
Certificate R6	direct scan through $q = 16$; finite bridge below 29	independent syndrome replay and separate radius gate
Certificate R6-NF	small exceptional normal forms	deterministic normal-form reconstruction; not a second implementation
Certificate R7	$q = 7, 8, 9, 11$ and finite bridge below 37	independent five-secant/orbit checks; split-free and radius flags remain separate
Certificate SC	integral stable-component model in every characteristic	exact polynomial identities and finite-field factorization replay; Singular checks saturation and vertical primary decompositions

Table 7 is a deterministic projection of the hash-pinned public R6 and R6-NF records. The top-level verifier regenerates it and checks byte-for-byte equality with the included source.

The public labels are defined by `supplement/CERTIFICATE-SCHEMA.md`. The paper-local bundle, literal replay commands, toolchain locks, hashes, byte counts, and working directories are recorded in `supplement/REPRODUCING.md`, `supplement/EVIDENCE-MANIFEST.json`, and `supplement/RELEASE-MANIFEST.md`. The single entry point `python3 supplement/verify.py` checks the bundle manifest, classification records and their expected hashes, and the local PDF and supplement-artifact rows printed in the release manifest; its `--replay` option runs the paper-local Python replays.

The paper-facing Lean gate checks coordinate algebra and conditional synthesis, not the geometric classifications or rational-point arguments. The exact declarations, hypotheses, axiom audit, and reproduction command are listed in `supplement/LEAN-STATEMENTS.md`. Lean checks the finite recursive descent from a classified bottom component and the simultaneous stagewise parameter-budget arithmetic. It proves polynomial density of monic split-squarefree coefficient tuples over an infinite field, the algebraic-closure setting used in the manuscript, and checks that containment of a dense attainable locus in a closed bad carrier extends to its rowspace closure and that an irreducible rowspace covered by finitely many closed components lies in one of them. Identification of those coefficient tuples with the retained-marker map, the concrete catalecticant-rowspace closure equality, properness and geometric integrality of the schemes, and the exact primary-decomposition ledger remain manuscript or certificate mathematics. Lean also checks that the uniform threshold lies in the Seroussi–Roth dimension range and that $(q, r) = (8, 6)$ is its exact endpoint. The formal adapter keeps the dual-GRS identification, Seroussi–Roth nonextendability, Dür equivalence, and the concrete radius proposition as separate hypotheses. The balanced quantum corollary has a separate cross-paper gate. Lean checks its field-eight certificate arithmetic, balanced-row uniqueness, and the length-ten LU and transversal-Clifford specializations; the standard MDS–AME and one-party Choi correspondences remain explicit cited inputs.

Data, code, and disclosure

The electronic supplement accompanying this manuscript contains the source, canonical classification representatives and stabilizers, certificate schemas, generators, independent replays where available, expected hashes, toolchain locks, and a top-level verification command. Section 7 assigns each computational or formal claim its exact trust route.

8 Scope and open problems

The contained/transverse fork gives a uniform all-level theorem. Theorem 1.3 supplies the exclusion theorem for reduced irreducible components without a field-size hypothesis, and Theorem 5.20 supplies the arithmetic threshold.

Corollary 8.1 (Large-characteristic stable list)**.** Let $r \geq 6$ and $q \geq Q_r$, and suppose $\mathcal{M}_r$ is empty. Then the split-free directions and the projective deep-hole directions are exactly the persistent tangent and conjugate-secant families, of total cardinality

$$\frac{q(q+1)^2}{2}.$$

The modular locus is empty when $\operatorname{char}\mathbb{F}_q > r-1$, so the conclusion applies in particular over every prime field with $q \geq Q_r$.

Proof. Theorem 5.20 leaves $\mathcal{P}_r \cup \mathcal{M}_r$, and the base-$p$ binomial criterion for normal-rational-curve nuclei [GH00, Lem. 1 and Thm. 1] makes $\mathcal{M}_r$ empty. The persistent families are split-free, and the Seroussi–Roth–Dür argument following Theorem 5.20 supplies covering radius $r-1$ throughout $q \geq Q_r$. The tangent family has $q(q+1)$ directions and the conjugate-secant family has $q(q^2-1)/2$, as in the uniform rank-two parametrization, giving the displayed total. □

Proposition 6.7 and Corollary 6.14 prove the first two assertions unconditionally. The characteristic hypothesis removes the modular term, not the large-characteristic R5 cyclic carrier; the theorem eliminates that carrier on the transverse branch.

Without the hypothesis $\mathcal{M}_r = \varnothing$, Theorems 1.3 and 5.20 give containment in $\mathcal{P}_r \cup \mathcal{M}_r$, not an exact classification within the modular locus.

1. Redundancy five is complete for every $q \geq 7$.
2. Redundancy six is a complete all-field deep-hole classification. Redundancy seven has a complete all-field split-free classification; its $q = 7, 8, 9$ covering-radius gate remains separate.
3. At every redundancy $r \geq 6$ and $q \geq Q_r$, every split-free direction, equivalently every projective deep-hole direction, lies in the persistent or modular locus. The binary cyclic-plane descendant terminates after redundancy seven, but fresh higher Lucas carriers remain possible. Proposition 5.22 gives the exact divisibility law on their canonical endpoint sections and eliminates the first fresh endpoint orbit. Fields below Q_r and exact arithmetic membership in the other modular-carrier strata remain separate.

In particular, the paper does not settle the general Reed–Solomon deep-hole conjecture: it neither classifies the fields below the transverse threshold nor determines exact membership in every higher modular-carrier stratum.

The last sporadic fields at redundancies five, six, and seven are $19, 13, 11$, while the corresponding point-count gates are $23, 29, 37$. These opposite trends ask whether sufficiently high redundancy has any sporadic fields outside the persistent and modular loci.

The structural conclusion is nevertheless stable. Beyond redundancy four, deepness is no longer controlled by one invariant or by isolated bad fibers. The persistent geometry is a coherent rank-two polar flag; modular geometry appears through characteristic-specific contraction kernels; and the remaining arithmetic is the Frobenius action on a normalized ordered-root cover. Redundancy five is the first exceptional-cover case, while redundancies six and seven show how that cover is propagated or contained.

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